

AN ESTIMATE FOR THE NUMBER OF TERMS IN THE HILBERT-KAMKE PROBLEM

UDC 511

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ABSTRACT. Let $r(n)$ denote the smallest s for which the system of equations

$$x_1^j + \dots + x_s^j = N_j \quad (j = 1, \dots, n) \quad (1)$$

is solvable in nonnegative integers $x_1, \dots, x_s$ for all sufficiently large natural numbers $N_1, \dots, N_n$ which satisfy the following conditions:

- 1) The singular integral $\gamma = \gamma(N_1, \dots, N_n)$ of the system (1) satisfies the inequality $\gamma \geq c(n, s) > 0$ (the order conditions).
- 2) The system of equations $\sum_{k=1}^n k^j t_k = N_j$ ($j = 1, \dots, n$) is solvable in integers $t_1, \dots, t_n$ (the arithmetic conditions).

In 1937, K. K. Mardzhanishvili proved that $n^2 \ll r(n) \leq n^4 2^{n^2-n-2}$. G. I. Arkhipov has recently obtained upper and lower estimates for $r(n)$ having the same order of magnitude: $2^n - 1 \leq r(n) \leq 3n^3 2^n - n$ ($n \geq 5$).

In this paper, the upper estimate for $r(n)$ is reduced to

$$r(n) \leq \sum_{0 \leq k \leq \lfloor \ln n / \ln 2 \rfloor} 2^k (2^{\lfloor n/2^k \rfloor} - 1) \quad (n \geq 12); \quad (2)$$

in particular, the asymptotic formula $r(n) = 2^n + O(2^{n/2})$ is obtained. It is conjectured that the estimate (2) is best possible.

Bibliography: 20 titles.

In this paper, building upon work of Arkhipov [1], [2], we continue to investigate the Hilbert-Kamke problem.

The Hilbert-Kamke problem is the study of the solvability of the system of Diophantine equations

$$x_1^j + \dots + x_s^j = N_j \quad (j = 1, \dots, n) \quad (1)$$

for all natural numbers $N_1, \dots, N_n$ which satisfy some rather general conditions and for s satisfying a bound depending on n . This problem, which was posed by Hilbert in 1909, was first mentioned in Kamke's paper [3], in which, under some general conditions on $N_1, \dots, N_n$ (concerning order of magnitude and divisibility), it is proved that (1) is solvable for a bounded number of variables. Kamke used a method which had been used by Hilbert (see [4]) to give a solution to Waring's problem for the polynomial x^n . As in the

1980 *Mathematics Subject Classification* (1985 Revision). Primary 11P05, 11D72; Secondary 11P55, 11D41, 11L03.

case of Waring's problem, the estimate for the number of terms s that is obtained using Hilbert's method is very large.

Further progress on the Hilbert-Kamke problem was connected with Vinogradov's method (see [5]–[7]). Using this method, Mardzhanishvili [8], [9] obtained for $s \geq cn^2 \ln n$ the following asymptotic formula for the number J of solutions of the system (1) with $1 \leq x_i \leq P$ ($i = 1, \dots, s$), where $P = N_n^{1/n}$:⁽¹⁾

$$J = \sigma \gamma P^{s-n(n+1)/2} + O(P^{s-n(n+1)/2-\rho}), \quad (2)$$

where $\rho > 0$ depends only on n and $\gamma = \gamma(N_1, \dots, N_n) \geq c_1(n, s) > 0$, if for some real numbers $I_j > 1$ and $i_j < 1$, $I_j < i_j s^{1-j/n}$ ($1 \leq j < n$), depending on s the natural numbers $N_1, \dots, N_n$ satisfy the conditions

$$I_j P^j \leq N_j \leq i_j s^{1-j/n} P^j \quad (j = 1, \dots, n-1), \quad (3)$$

and $\sigma = \sigma(N_1, \dots, N_n) \geq c_2(n, s) > 0$, if $s \geq 2^{n^2-n-2} n^4$, $n \geq 2$; and if n congruences hold, congruences which are (as Arkhipov noticed) equivalent to solvability in integers $t_1, \dots, t_n$ of the system of linear equations $\sum_{k=1}^n k^j t_k = N_j$ ($j = 1, \dots, n$). Mardzhanishvili also proved that his arithmetic conditions on $N_1, \dots, N_n$, given in the form of congruences, are necessary conditions for (1) to be solvable. The order conditions (3) were used even by Kamke. They are close to the necessary conditions $N_n^{j/n} \leq N_j \leq s^{1-j/n} N_n^{j/n}$ ($j = 1, \dots, n-1$). We note that, under the same conditions on $N_1, \dots, N_n$ that Mardzhanishvili has, Emel'yanov [10] gave an elementary proof that system (1) with a bounded number of terms is solvable for P sufficiently large.

Until recently it was conjectured (see [7], Russian p. 12) that, if $N_1, \dots, N_n$ satisfy order conditions⁽²⁾ and arithmetic conditions, then system (1) becomes solvable for N sufficiently large when s has order n^2 . This conjecture was disproved by Arkhipov [2], who found the correct order for s .

Let $r(n)$ be the smallest natural number s such that, if one has order conditions which ensure the inequality $\gamma \geq c_1(n, s) > 0$, and the necessary arithmetic conditions on $N_1, \dots, N_n$, then (1) is solvable for N_n sufficiently large. From results of Arkhipov it follows that $2^n - 1 \leq r(n) \leq 3n^3 2^n - n$ ($n \geq 5$). Arkhipov also found necessary and sufficient conditions for $\gamma > 0$. In particular, he showed that, if $\gamma = 0$ and $s > \frac{1}{2}n(n+1) + 1$, then system (1) has at most $c(n, s)$ solutions, and may have no solutions. The same can occur when γ is greater than 0 but is too small in comparison with the order of magnitude of N_n . This justifies the imposition of order conditions on $N_1, \dots, N_n$ with $\gamma \geq c_1(n, s) > 0$.

In the present paper we prove that the inequality $r(n) \leq H(n)$ holds for $n \geq 12$, where

$$H(n) = \sum_{0 \leq k \leq [\ln n / \ln 2]} 2^k (2^{\lfloor n/2^k \rfloor} - 1).$$

The proof of Proposition 2 can be used to obtain a lower bound for $r(n)$. Here, using induction, one easily obtains the bound $r(n) \geq F(n)$, where $F(n)$ is defined recursively by means of the formulas $F(0) = 0$ and $F(k) = 2^k - 1 + 2F(\lfloor (k-1)/2 \rfloor)$ ($k \geq 1$); in

⁽¹⁾ See the definition of the singular series σ and the singular integral γ in [2], §1.

⁽²⁾ By "order conditions" we mean conditions on $N_1, \dots, N_n$ which ensure that (1) has real solutions in intervals of any fixed length.

particular, for $n = 2^m - 1$ we obtain the equality $r(n) = H(n)$. We conjecture that this equality $r(n) = H(n)$ holds for $n \geq 12$. In a later publication we plan to give a proof of a lower bound for $r(n)$ which is necessary for this. The method of the present article also enables one to find the exact values of $r(n)$ for $n < 12$.

Notation. v, i, j, k, l , and m are nonnegative integers; q, n , and s are natural numbers; $[\alpha]$ denotes the integer part and $\{\alpha\}$ the fractional part of α ; p is a prime number; and

$$p^k \| a - p^k | a \wedge p^{k+1} \nmid a; \quad \Delta_h^{(0)} g(x) = g(x), \quad \Delta_h g(x) = g(x+h) - g(x),$$

$$\Delta_h^{(v+1)} g(x) = \Delta_h(\Delta_h^{(v)} g(x)) \quad (v = 0, 1, \dots);$$

$$\binom{x}{0} = 1, \quad \binom{x}{k} = \frac{x(x-1) \cdots (x-k+1)}{k!} \quad (k = 1, 2, \dots).$$

§1. Lemmas

We say that a polynomial $\varphi(x)$ is *integer-valued* if $\varphi(x)$ is an integer whenever x is an integer.

LEMMA 1. *Let $f(x)$ be a polynomial of degree n with real coefficients. For $f(x)$ to be integer-valued it is necessary and sufficient that it be expressible in the form*

$$c_n \binom{x}{n} + \cdots + c_0 \binom{x}{0},$$

where $c_n, \dots, c_0$ are integers.

For the proof, see [11], Chapter 1.

LEMMA 2. *Let h be a real number, and let $g(x)$ be an arbitrary function on the set of all real numbers. Then*

$$\Delta_{kh} g(x) = \sum_{i=1}^k \Delta_h^{(i)} g(x) \binom{k}{i}.$$

For the proof, see [12], Chapter 1, §3.3.

COROLLARY 1. *Let*

$$f(x) = a_n \binom{x}{n} + \cdots + a_0 \binom{x}{0}.$$

Then

$$\Delta_m f(x) = \sum_{j=0}^{n-1} \binom{x}{j} \sum_{v=1}^{n-j} \binom{m}{v} a_{v+j}.$$

PROOF. If we note that

$$\Delta_1^{(v)} \binom{x}{i} = \binom{x}{i-v} \quad (i \geq v)$$

and apply Lemma 2, we obtain the required equality:

$$\begin{aligned} \Delta_m f(x) &= \sum_{v=1}^m \Delta_1^{(v)} f(x) \binom{m}{v} = \sum_{v=1}^m \sum_{i=v}^n a_i \binom{x}{i-v} \binom{m}{v} \\ &= \sum_{j=0}^{n-1} \binom{x}{j} \sum_{v=1}^{\min(n-j, m)} \binom{m}{v} a_{v+j} = \sum_{j=0}^{n-1} \binom{x}{j} \sum_{v=1}^{n-j} \binom{m}{v} a_{v+j}. \end{aligned}$$

COROLLARY 2. Let h be a real number, and let $f(x)$ be a polynomial of degree n . Then

$$\Delta_{hy}f(x) = \sum_{\nu=1}^n \Delta_h^{(\nu)}f(x) \binom{y}{\nu}. \quad (4)$$

PROOF. For a natural number $y \geq n$ equality (4) follows from Lemma 2. But since both sides of (4) are polynomials in y , this equality must hold identically.

LEMMA 3. Suppose that k, m , and ν are natural numbers, $\kappa \geq 0$ is an integer, $p^k \mid m$ and $p^\kappa \parallel \nu$. Then $\binom{m}{\nu} \equiv 0 \pmod{p^{\max(0, k-\kappa)}}$.

PROOF. We suppose that $\kappa < k$ and $\nu < m$, since otherwise the lemma is obvious. The exponent of p which occurs in the prime factorization of $\binom{m}{\nu}$ is equal to

$$\sum_{i=1}^{\infty} \left(\left\lfloor \frac{m}{p^i} \right\rfloor - \left\lfloor \frac{\nu}{p^i} \right\rfloor - \left\lfloor \frac{m-\nu}{p^i} \right\rfloor \right) \geq \sum_{i=\kappa+1}^k \left(\left\lfloor \frac{m}{p^i} \right\rfloor - \left\lfloor \frac{\nu}{p^i} \right\rfloor - \left\lfloor \frac{m-\nu}{p^i} \right\rfloor \right).$$

Since $\kappa + 1 \leq i \leq k$ implies that $p^i \nmid \nu$ and $p^i \nmid m - \nu$, and since m is divisible by p^i , it follows that

$$\left\lfloor \frac{m}{p^i} \right\rfloor - \left\lfloor \frac{\nu}{p^i} \right\rfloor - \left\lfloor \frac{m-\nu}{p^i} \right\rfloor = 1,$$

and hence $p^{k-\kappa} \mid \binom{m}{\nu}$. The lemma is proved.

COROLLARY. Suppose that k is a natural number, $t = [\ln k / \ln p]$, and $\varphi(x)$ is an integer-valued polynomial of degree k . Then, for any natural number l and all integers x , $\Delta_{p^{l+t}}\varphi(x) \equiv 0 \pmod{p^l}$.

PROOF. According to Lemma 2,

$$\Delta_{p^{l+t}}\varphi(x) = \sum_{\nu=1}^k \Delta_1^{(\nu)}\varphi(x) \binom{p^{l+t}}{\nu},$$

at the same time as Lemma 3 gives us $\binom{p^{l+t}}{\nu} \equiv 0 \pmod{p^l}$, and so the required congruence follows.

LEMMA 4. Let $\varphi(x)$ be an integer-valued polynomial of degree $n-1$, let k be a natural number, and let $\beta \geq [(n-1)/(p-1)]$ be an integer. There exists an integer-valued polynomial

$$f(x) = a_n \binom{x}{n} + \cdots + a_1 \binom{x}{1}$$

such that

$$\Delta_{p^k}f(x) = p^{\beta+k}\varphi(x). \quad (5)$$

The coefficients of $f(x)$ satisfy the congruences

$$a_{n-i} \equiv 0 \pmod{p^{\beta - [i/(p-1)]}} \quad (i = 0, \dots, n-1). \quad (6)$$

PROOF. In accordance with Lemma 1, we represent $\varphi(x)$ in the form: $\varphi(x) = b_{n-1} \binom{x}{n-1} + \cdots + b_0 \binom{x}{0}$, where $b_{n-1}, \dots, b_0$ are integers. By Corollary 1 of Lemma 2, relation (5) is equivalent to

$$\sum_{\nu=1}^{n-j} \binom{p^k}{\nu} a_{\nu+j} = p^{\beta+k} b_j \quad (j = 0, \dots, n-1). \quad (7)$$

This system has a triangular matrix relative to the $a_n, \dots, a_1$ with diagonal entries equal to p^k ; hence, the numbers $a_n, \dots, a_1$ are uniquely determined. We now use induction on i to prove that a_{n-i} is an integer satisfying (6).

For $j = n - 1$ it follows from (7) that $p^k a_n = p^{\beta+k} b_{n-1}$, from which we have $a_n = p^\beta b_{n-1}$, and (6) holds with $i = 0$. Suppose that the a_{n-i} are integers satisfying (6) for all i in the interval $0 \leq i < m$, where $m \leq n - 1$. We write (7) with $j = n - 1 - m$ in the form

$$p^k a_{n-m} = - \sum_{i=0}^{m-1} \binom{p^k}{m+1-i} a_{n-i} + p^{\beta+k} b_{n-1-m}.$$

According to Lemma 3 and the induction assumption, the right side is divisible by p^λ , where

$$\begin{aligned} \lambda &= \min \left(\beta + k, \min_{0 \leq i \leq m-1} \left(\beta - \left\lfloor \frac{i}{p-1} \right\rfloor + \max \left(0, k - \left\lfloor \frac{\ln(m+1-i)}{\ln p} \right\rfloor \right) \right) \right) \\ &\geq \beta + k - \max_{0 \leq i \leq m-1} \left(\left\lfloor \frac{i}{p-1} \right\rfloor + \left\lfloor \frac{m-i}{p-1} \right\rfloor \right) = \beta + k - \left\lfloor \frac{m}{p-1} \right\rfloor \geq k. \end{aligned}$$

Consequently, a_{n-m} is an integer, and (6) holds for $i = m$. This completes the proof of the lemma by induction.

LEMMA 5. Suppose that $a_j(x_1, \dots, x_{j-1})$ and $\varphi_j(x_1, \dots, x_n)$ ($j = 1, \dots, n$) are polynomials with p -adic integer coefficients. Then the system of equations

$$x_j = a_j(x_1, \dots, x_{j-1}) + p\varphi_j(x_1, \dots, x_n) \quad (j = 1, \dots, n) \quad (8)$$

is solvable in p -adic integers $x_1, \dots, x_n$.

PROOF. We consider the system of congruences ($j = 1, \dots, n$)

$$x_j \equiv a_j(x_1, \dots, x_{j-1}) + p\varphi_j(x_1, \dots, x_n) \pmod{p^k}. \quad (9)$$

When $k = 1$ this system is obviously solvable. Suppose that $(x_1^{(m)}, \dots, x_n^{(m)})$ is a solution when $k = m$. We shall construct a solution $(x_1^{(m+1)}, \dots, x_n^{(m+1)})$ to (9) with $k = m + 1$ in the following form: $x_j^{(m+1)} = x_j^{(m)} + p^m u_j$ ($j = 1, \dots, n$), where the u_j are integers.

Since for any integers u_j we have

$$\varphi_j(x_1^{(m+1)}, \dots, x_n^{(m+1)}) \equiv \varphi_j(x_1^{(m)}, \dots, x_n^{(m)}) \pmod{p^m},$$

it follows that we can ensure that $x_1^{(m+1)}, \dots, x_n^{(m+1)}$ satisfy (9) with $k = m + 1$ by choosing $u_1, \dots, u_n$ so as to satisfy the congruences

$$\begin{aligned} x_j^{(m)} + p^m u_j &\equiv a_j(x_1^{(m)} + p^m u_1, \dots, x_{j-1}^{(m)} + p^m u_{j-1}) \\ &\quad + p\varphi_j(x_1^{(m)}, \dots, x_n^{(m)}) \pmod{p^{m+1}} \quad (j = 1, \dots, n). \end{aligned}$$

This is possible, because, by assumption,

$$x_j^{(m)} \equiv a_j(x_1^{(m)}, \dots, x_{j-1}^{(m)}) + p\varphi_j(x_1^{(m)}, \dots, x_n^{(m)}) \pmod{p^m}.$$

By induction, we obtain a sequence $(x_1^{(k)}, \dots, x_n^{(k)})$ ($k = 1, 2, \dots$) of solutions to (9), and the sequence satisfies the condition $x_j^{(k+1)} \equiv x_j^{(k)} \pmod{p^k}$ ($j = 1, \dots, n$). The latter condition implies that the sequence $(x_1^{(k)}, \dots, x_n^{(k)})$ ($k = 1, 2, \dots$) converges in the p -adic metric to an n -tuple $(\alpha_1, \dots, \alpha_n)$ of p -adic integers. Using the continuity of polynomials, we see that $\alpha_1, \dots, \alpha_n$ satisfy (8). The lemma is proved.

LEMMA 6. Let $n \geq 2$ and $p > n$. The system of congruences

$$x_1^j + \dots + x_s^j \equiv N_j \pmod{p} \quad (j = 1, \dots, n) \quad (10)$$

is solvable for $s \geq \min((p-1)n, [2en^2 \ln n])$ and for any integers $N_1, \dots, N_n$.

PROOF. The solvability of (10) when $s \geq (p-1)n$ is proved as a special case in Lemma 6 of [13]. We set $h = [en]$, $k = [\ln(n! - 1)] + 1$ and consider the system of congruences

$$\sum_{i=1}^{2n} y_i^j + \sum_{v=1}^k (u_v + v_v) x_v^j \equiv N_j \pmod{p} \quad (j = 1, \dots, n),$$

where $1 \leq x_i$, $y_i \leq p$ and $0 \leq u_v$, $v_v \leq h$. We write the number T of solutions of this system in terms of trigonometric sums:

$$T = p^{-n} \sum_{a_1, \dots, a_n=0}^{p-1} \left(\sum_{y=1}^p e^{2\pi i f(y)/p} \right)^{2n} \left(\sum_{x=1}^p \sum_{u, v=0}^h e^{2\pi i (u+v)f(x)/p} \right)^k e^{-2\pi i \lambda/p},$$

where $f(x) = a_1 x + \dots + a_n x^n$ and $\lambda = a_1 N_1 + \dots + a_n N_n$. Since a polynomial $f(x)$ with $(a_1, \dots, a_n, p) = 1$ takes each value modulo p at most n times when x runs through the integers in the interval $[1, p]$, it follows that when $(a_1, \dots, a_n, p) = 1$ we have

$$\left| \sum_{x=1}^p \sum_{u, v=0}^h e^{2\pi i (u+v)f(x)/p} \right| \leq n \sum_{a=0}^{p-1} \left| \sum_{u, v=0}^h e^{2\pi i (u+v)a/p} \right| = np(h+1).$$

From this we obtain

$$\begin{aligned} T &\geq (h+1)^{2k} p^{k+n} - (np(h+1))^k \left(p^{-n} \sum_{a_1, \dots, a_n=0}^{p-1} \left| \sum_{y=1}^p e^{2\pi i f(y)/p} \right|^{2n} - p^n \right) \\ &= (h+1)^{2k} p^{k+n} - (np(h+1))^k (T_0 - p^n), \end{aligned}$$

where T_0 is the number of solutions of the system of congruences $x_1^j + \dots + x_n^j \equiv y_1^j + \dots + y_n^j \pmod{p}$ ($j = 1, \dots, n$). Mordell [14] showed that T_0 satisfies the bound $T_0 \leq n!p^n$. Consequently,

$$T \geq (h+1)^k p^{k+n} ((h-1)^k - n^k(n! - 1)) > 0.$$

Since $T > 0$, it follows that (10) is solvable when $s \geq 2n + 2kh$. We now obtain the lemma, using the inequalities $2n + 2kh \leq 2n[2en(\ln(n! - 1) + 1)] \leq [2en^2 \ln n]$.

REMARK. The proof of Lemma 6 combines the method of Karatsuba in [15] with the Mordell estimate $T_0 \leq n!p^n$. Here, unlike in [15] (see also [16] and [13]), we did not use the estimate of Weil [17] for rational trigonometric sums with prime denominator.

LEMMA 7. Let $p-1 \mid n$ and $\beta = n/(p-1)$. The polynomial

$$f_n(x) = \sum_{i=1}^n a_i \binom{x}{i}, \quad \text{where } a_i = -\frac{1}{p} \sum_{a=1}^p (e^{2\pi i a/p} - 1)^i,$$

is integer-valued and satisfies the following congruences for x an integer:

$$f_n(x) \equiv \begin{cases} 0 \pmod{p^\beta}, & \text{if } x \equiv 0 \pmod{p}, \\ 1 \pmod{p^\beta}, & \text{if } x \not\equiv 0 \pmod{p}. \end{cases}$$

The coefficients of $f_n(x)$ satisfy the conditions

$$p^{\beta-1} \parallel a_n, \quad a_{n-j} \equiv 0 \pmod{p^{\beta-1-[j/(p-1)]}} \quad (j = 1, \dots, n-1).$$

PROOF. We consider the function

$$g(x) = 1 - \frac{1}{p} \sum_{a=1}^p e^{2\pi i a x/p} = \begin{cases} 0, & \text{if } x \equiv 0 \pmod{p}; \\ 1, & \text{if } x \not\equiv 0 \pmod{p}. \end{cases}$$

If x is a natural number, then

$$\begin{aligned} g(x) &= 1 - \frac{1}{p} \sum_{a=1}^p (1 + e^{2\pi i a/p} - 1)^x = 1 - \frac{1}{p} \sum_{a=1}^p \sum_{j=0}^x \binom{x}{j} (e^{2\pi i a/p} - 1)^j \\ &= -\frac{1}{p} \sum_{j=1}^p \binom{x}{j} \sum_{a=1}^p (e^{2\pi i a/p} - 1)^j = \sum_{j=1}^x a_j \binom{x}{j}, \end{aligned} \quad (11)$$

where

$$\begin{aligned} a_j &= -\frac{1}{p} \sum_{a=1}^p (e^{2\pi i a/p} - 1)^j = -\frac{1}{p} \sum_{l=0}^j (-1)^{j-l} \binom{j}{l} \sum_{a=1}^p e^{2\pi i a l/p} \\ &= \sum_{0 \leq k \leq \lfloor j/p \rfloor} (-1)^{j-kp+1} \binom{j}{kp} \end{aligned}$$

are integers. In the cyclotomic extension $\mathbf{Q}(e^{2\pi i/p})$ of the field of rational numbers $\mathbf{Q}$, the number $\lambda = 1 - e^{2\pi i/p}$ is a prime element of norm p (see [18], Chapter 3, §1 and Chapter 5, §2). Since λ^j divides pa_j in the ring of algebraic integers of $\mathbf{Q}(e^{2\pi i/p})$, by passing to norms we see that $p^j \mid p^{p-1}a_j^{p-1}$. Consequently, a_j is divisible by p^{δ_j} , where δ_j is the least integer for which $\delta_j \geq j/(p-1) - 1$. If $0 \leq \nu \leq n-1$, then $\delta_{n-\nu} = \beta - 1 - \lfloor \nu/(p-1) \rfloor$, and when $j > n$ we obtain the congruence $a_j \equiv 0 \pmod{p^\beta}$.

Suppose that $x \geq n$. Passing to congruence modulo p^β in (11) gives $g(x) \equiv f_n(x) \pmod{p^\beta}$. Because $f_n(x)$ and $g(x)$ are periodic mod p^β , this congruence holds for all integers x . Thus, it remains to prove that $p^{\beta-1} \parallel a_n$. We have

$$\frac{1}{\lambda} (e^{2\pi i a/p} - 1) = - \sum_{\nu=0}^{a-1} e^{2\pi i \nu/p} \equiv -a \pmod{\lambda},$$

and, since $\lambda \mid p$, it follows that

$$\lambda^{-n} p a_n \equiv - \sum_{a=1}^{p-1} (-a)^n \equiv 1 \pmod{\lambda}.$$

Since λ^{p-1} and p are associates, it now follows that $p^{\beta-1} \parallel a_n$. The lemma is proved.

REMARK. When $p = 2$ the polynomial $f_n(x) = \sum_{i=1}^n (-2)^{i-1} \binom{x}{i}$ was defined by Hua [19]. A polynomial with properties analogous to $f_n(x)$ was used by Arkhipov ([2], Lemmas 16 and 17) in order to obtain the estimate $r(n) \geq 2^n - 1$.

For $f_n(x)$ and $p = 2$, according to Lemma 2 we have

$$\begin{aligned} \Delta_2 f_n(x) &= \sum_{\nu=1}^2 \Delta_1^{(\nu)} f_n(x) \binom{2}{\nu} \\ &= \sum_{i=1}^n (-2)^{i-1} \binom{x}{i-1} 2 + \sum_{i=2}^n (-2)^{i-1} \binom{x}{i-2} \\ &= (-1)^{n-1} 2^n \binom{x}{n-1}. \end{aligned}$$

LEMMA 8. Let $f(x)$ be an integer-valued polynomial of degree n . Then, for any natural numbers m and v and any integer x ,

$$\Delta_{p^m}^{(v)} f(x) \equiv 0 \pmod{p^\gamma}, \quad \text{where } \gamma = \max\left(0, mv - \left\lfloor \frac{n-v}{p-1} \right\rfloor - \dots - \left\lfloor \frac{n-v}{p^m-1} \right\rfloor\right).$$

PROOF. By Lemma 1, we have

$$\Delta_{p^m}^{(v)} f(x) = \sum_{\substack{1 \leq i_1, \dots, i_v \leq p^m \\ i_1 + \dots + i_v \leq n}} \Delta_1^{(i_1 + \dots + i_v)} f(x) \binom{p^m}{i_1} \dots \binom{p^m}{i_v}.$$

Let h_j be the number of integers $i_1, \dots, i_v$ which are divisible by p^j . Then $v - h_j + p^j h_j \leq n$, and hence $h_j \leq [(n-v)/(p^j-1)]$. Applying Lemma 3, we find that the number $\binom{p^m}{i_1} \dots \binom{p^m}{i_v}$ is divisible by p to the power

$$\max(0, mv - ((h_1 - h_2) + \dots + (m-1)(h_{m-1} - h_m)) + mh_m) \geq \gamma.$$

Consequently, $\Delta_{p^m}^{(v)} f(x)$ is divisible by p^γ . The lemma is proved.

LEMMA 9. Let $s \geq \frac{1}{2}n(n+1) + 2$. Then the singular series σ in (2) satisfies the relation

$$\sigma = \prod_p \varphi_p, \quad \text{where } \varphi_p = \lim_{k \rightarrow \infty} W(p^k, s) p^{k(n-s)},$$

and $W(p^k, s)$ is the number of solutions of the system of congruences

$$\sum_{i=1}^s x_i^j \equiv N_j \pmod{p^k} \quad (j = 1, \dots, n).$$

For the proof, see [2], Lemma 9.

LEMMA 10. Suppose that $n, m_1, \dots, m_n$ are natural numbers with $|m_i - m_j| \leq 1$ ($i, j = 1, \dots, n$), and the a_{ij} ($i, j = 1, \dots, n$) are p -adic integers satisfying $a_{j,j} \not\equiv 0 \pmod{p}$ and $a_{i,j} \equiv 0 \pmod{p}$ for $i < j$. Then the system of congruences

$$\sum_{j=1}^n a_{i,j} x_j \equiv A_i \pmod{p^{m_i}} \quad (i = 1, \dots, n) \quad (12)$$

is solvable in integers $x_1, \dots, x_n$ in the intervals $0 \leq x_j \leq p^{m_j} - 1$ ($j = 1, \dots, n$) for any p -adic integers $A_1, \dots, A_n$.

PROOF. We set $m = \min(m_1, \dots, m_n)$. Then $m_i \leq m + 1$ ($i = 1, \dots, n$). Since

$$\det((a_{i,j})_{j=1, \dots, n}^{i=1, \dots, n}) \not\equiv 0 \pmod{p},$$

it follows that there exist integers $x_j^{(0)}$, $0 \leq x_j^{(0)} \leq p^m - 1$ ($j = 1, \dots, n$), which satisfy the system of congruences $\sum_{j=1}^n a_{i,j} x_j^{(0)} \equiv A_i \pmod{p^m}$ ($i = 1, \dots, n$). Let $i_1 < \dots < i_k$ be all of the indices i for which $m_i = m + 1$. We set $x_{i_l} = x_{i_l}^{(0)} + p^m t_l$ and $x_j = x_j^{(0)}$, if $j \neq i_l$ ($l = 1, \dots, k$). To prove that (12) is solvable in the intervals $0 \leq x_j \leq p^{m_j} - 1$ ($j = 1, \dots, n$), it is sufficient to prove that the system

$$\sum_{l=1}^k a_{i_s, i_l} t_l \equiv B_s \pmod{p} \quad (s = 1, \dots, k)$$

is solvable in integers $t_1, \dots, t_k$ with $0 \leq t_l \leq p-1$ ($l = 1, \dots, k$) for certain integers $B_1, \dots, B_k$. But this follows because $a_{i_s, i_s} \not\equiv 0 \pmod{p}$ ($s = 1, \dots, k$). The lemma is proved.

LEMMA 11. *The asymptotic formula (2) holds for $s \geq n^2(4 \ln n + 2 \ln \ln n + 9)$.*

For the proof see [2], Theorem 1.

§2. Preliminary estimates

Let

$$\binom{x}{j} = \sum_{\nu=1}^j \mu_{j,\nu} x^\nu \quad (j = 1, \dots, n), \quad x^j = \sum_{\nu=1}^j \theta_{j,\nu} \binom{x}{\nu} \quad (j = 1, \dots, n),$$

where, by Lemma 1, the $\theta_{j,\nu}$ are integers. If for some integers $A_1, \dots, A_n$ the system of equations

$$\sum_{i=1}^s \binom{x_i}{j} = A_j \quad (j = 1, \dots, n) \quad (13)$$

is solvable in integers $x_1, \dots, x_s$, then for the integers $N_1, \dots, N_n$ which are connected to $A_1, \dots, A_n$ by the relations

$$N_j = \sum_{\nu=1}^j \theta_{j,\nu} A_\nu \quad (j = 1, \dots, n), \quad (14)$$

the system of equations

$$\sum_{i=1}^s x_i^j = N_j \quad (j = 1, \dots, n) \quad (15)$$

is solvable in integers $x_1, \dots, x_s$. Conversely, if (15) is solvable for certain integers $N_1, \dots, N_n$, then the numbers $A_1, \dots, A_n$ determined from (14) by the relations $A_j = \sum_{\nu=1}^j \mu_{j,\nu} N_\nu$ ($j = 1, \dots, n$) are integers, and (13) is solvable. It should be noted that (13) has an advantage over (15): in (13) it is not necessary to impose arithmetic solvability conditions on $A_1, \dots, A_n$, since the system of congruences $\sum_{i=1}^s \binom{x_i}{j} \equiv A_j \pmod{q}$ ($j = 1, \dots, n$) is solvable in integers $x_1, \dots, x_s$ for any integers $A_1, \dots, A_n$, whenever $s \geq (q-1)n$.

We let $r(n, p)$ denote the smallest natural number s such that (13) is solvable in p -adic integers $x_1, \dots, x_s$ for any integers $A_1, \dots, A_n$. We note that for $s \geq r(n, p)$ system (13) is also solvable in p -adic integers $x_1, \dots, x_s$ if $A_1, \dots, A_n$ are any p -adic integers. In fact, if $\bar{x}^{(j)} = (x_1^{(j)}, \dots, x_n^{(j)})$ is a solution corresponding to the integer n -tuple $(A_1^{(j)}, \dots, A_n^{(j)})$, and if the sequence $(A_1^{(j)}, \dots, A_n^{(j)})$ ($j = 1, 2, \dots$) converges to $(A_1, \dots, A_n)$ in the p -adic metric, then, by compactness, one can choose a subsequence of $\bar{x}^{(j)}$ ($j = 1, 2, \dots$) which, by the continuity of polynomials, converges to a solution which corresponds to $(A_1, \dots, A_n)$.

An n -tuple of p -adic integers $(\lambda_1, \dots, \lambda_n)$ will be called a *rank- n lifting tuple modulo p* if

$$\det \left(\binom{\lambda_i}{j} \right)_{j=0, \dots, n-1}^{i=1, \dots, n} = \prod_{1 \leq l < k \leq n} \frac{\lambda_k - \lambda_l}{k - l} \not\equiv 0 \pmod{p}.$$

An example of a rank- n lifting tuple is the set of numbers $\lambda_i = a + i$ ($i = 1, \dots, n$), where a is an integer.

We let $\bar{r}(n, p)$ denote the smallest natural number s such that for any integers $A_1, \dots, A_n$ system (13) is solvable in p -adic integers $x_1, \dots, x_s$ which include a rank- n lifting tuple modulo p . From the definition of $r(n, p)$ and $\bar{r}(n, p)$ it follows that $r(n, p) \leq \bar{r}(n, p) \leq r(n, p) + n$.

LEMMA 12. Suppose that $n \geq 2$, $\kappa = [\ln(n-1)/\ln p] + 1$, $(\lambda_1, \dots, \lambda_n)$ is a rank- n lifting tuple (mod p); $A_1, \dots, A_n$ are integers, and the system of congruences

$$\sum_{i=1}^n \binom{x_i}{j} \equiv A_j \pmod{p^{k+[(n-j)/(p-1)]}} \quad (j = 1, \dots, n) \quad (16)$$

is satisfied for some set of integers $x_1, \dots, x_n$ with $x_i \equiv \lambda_i \pmod{p^\kappa}$ ($i = 1, \dots, n$), and $k = \kappa + \delta$, where $\delta = 1 - 2\{p/2\}$. Then for any integer $k \geq \kappa + \delta$ the system of congruences (16) is solvable in integers $x_1, \dots, x_n$ with $x_i \equiv \lambda_i \pmod{p^\kappa}$ ($i = 1, \dots, n$).

PROOF. We prove the lemma by induction on $k \geq \kappa + \delta$. When $k = \kappa + \delta$ it is true by assumption. Suppose that for $k = m$, where $m \geq \kappa + \delta$, system (16) has a solution $x_1, \dots, x_n$ with $x_i \equiv \lambda_i \pmod{p^m}$ ($i = 1, \dots, n$). We shall construct a solution $x'_1, \dots, x'_n$ of (16) with $k = m + 1$ in the form $x'_i = x_i + p^m z_i$ ($i = 1, \dots, n$), where the z_i are integers.

By our assumption, we may let

$$A_j - \sum_{i=1}^n \binom{x_i}{j} = M_j p^{m+[(n-j)/(p-1)]} \quad (j = 1, \dots, n),$$

where the M_j are integers. Then the system of congruences

$$\sum_{i=1}^n \binom{x'_i}{j} \equiv A_j \pmod{p^{m+1+[(n-j)/(p-1)]}} \quad (j = 1, \dots, n)$$

is equivalent to the system

$$\sum_{i=1}^n \Delta_{p^m z_i} \binom{x_i}{j} \equiv M_j p^{m+[(n-j)/(p-1)]} \pmod{p^{m+1+[(n-j)/(p-1)]}} \quad (j = 1, \dots, n)$$

and, after using Corollary 2 of Lemma 2 to transform the left side, we see that it is equivalent to the system ($j = 1, \dots, n$)

$$\sum_{i=1}^n \left(\sum_{\nu=1}^j \Delta_{p^m}^{(\nu)} \binom{x_i}{j} \binom{z_i}{\nu} \right) \equiv M_j p^{m+[(n-j)/(p-1)]} \pmod{p^{m+1+[(n-j)/(p-1)]}}.$$

To prove that the last system of congruences is solvable in integers $z_1, \dots, z_n$, it suffices to show that the system of equations

$$\sum_{i=1}^n \left(\sum_{\nu=1}^j \Delta_{p^m}^{(\nu)} \binom{x_i}{j} \binom{z_i}{\nu} \right) = M_j p^{m+[(n-j)/(p-1)]} \quad (j = 1, \dots, n) \quad (17)$$

is solvable in p -adic integers $z_1, \dots, z_n$.

Using Lemma 4, we find integer-valued polynomials $f_l(x) = \sum_{j=1}^l a_{l,j} \binom{x}{j}$ ($l = 1, \dots, n$) which satisfy the relations

$$\Delta_{p^m} f_l(x) = p^{m+[(l-1)/(p-1)]} \binom{x}{l-1} \quad (l = 1, \dots, n). \quad (18)$$

Here the coefficients $a_{l,j}$ satisfy the conditions

$$a_{l,l} \neq 0, \quad a_{l,j} \equiv 0 \pmod{p^{[(l-1)/(p-1)] - [(l-j)/(p-1)]}}, \quad 1 \leq j \leq l \leq n. \quad (19)$$

For all l , $1 \leq l \leq n$, we form the linear combinations of the equations (17) ($j = 1, \dots, l$) with coefficients $a_{l,j}$. Then, taking (18) and (19) into account, we obtain the following system which is equivalent to (17):

$$\sum_{i=1}^n \left(\sum_{v=1}^l \Delta_{p^m}^{(v-1)} \binom{x_i}{l-1} \binom{z_i}{v} \right) = C_l \quad (l = 1, \dots, n), \quad (20)$$

where the integers C_l are determined by

$$C_l = p^{-m - [(l-1)/(p-1)]} \sum_{j=1}^l a_{l,j} M_j p^{m + [(n-j)/(p-1)]} \quad (l = 1, \dots, n).$$

Since $p^{\kappa+\delta}$ is a period modulo $p^{\delta+1}$ for all integer-valued polynomials of degree at most $n-1$, by the corollary to Lemma 3 it follows that for $v \geq 2$ we have the congruences

$$\Delta_{p^m}^{(v-1)} \binom{x_i}{l-1} \equiv 0 \pmod{p^{(v-1)(\delta+1)}} \quad (l, i = 1, \dots, n).$$

If α is the integer for which $p^\alpha \parallel v!$, we have $\alpha = \sum_{i=1}^\infty [\nu/p^i] < \nu/(p-1)$. This implies that in the coefficients of the powers of z in $\Delta_{p^m}^{(v-1)} \binom{x_i}{l-1} \binom{z_i}{v}$, where $v \geq 2$, p occurs to a power $> (v-1)(\delta+1) - \nu/(p-1) \geq 0$. The equations (20) can now be represented in the form

$$\sum_{i=1}^n \binom{x_i}{l-1} z_i = C_l + p \psi_l(z_1, \dots, z_n) \quad (l = 1, \dots, n), \quad (21)$$

where the $\psi_l(z_1, \dots, z_n)$ are polynomials with p -adic integer coefficients. Since $(\lambda_1, \dots, \lambda_n)$ is a rank- n lifting tuple $(\bmod p)$, we have

$$\det \left(\binom{x_i}{l-1} \right)_{l=1, \dots, n}^{i=1, \dots, n} = \prod_{1 \leq l < t \leq n} \frac{x_t - x_l}{t - l} \equiv \prod_{1 \leq l < t \leq n} \frac{\lambda_t - \lambda_l}{t - l} \not\equiv 0 \pmod{p}$$

and (21) is equivalent to a system of the form

$$z_i = a_i + p \varphi_i(z_1, \dots, z_n) \quad (i = 1, \dots, n), \quad (22)$$

where the a_i are p -adic integers and the $\varphi_i(z_1, \dots, z_n)$ are polynomials with p -adic integer coefficients.

Solvability of (22) in p -adic integers $z_1, \dots, z_n$ follows from Lemma 5. We have thus proved that system (16) with $k = m+1$ has a solution $x'_1, \dots, x'_n$. Since, by construction, $x'_i \equiv x_i \equiv \lambda_i \pmod{p^\kappa}$ ($i = 1, \dots, n$), the lemma holds for $k = m+1$. This completes the proof of the lemma by induction.

COROLLARY. Let $n \geq 2$. For $p > n$ one has

$$\bar{r}(n, p) \leq \min(pn, [2en^2 \ln n] + n),$$

and for $p \leq n$

$$\bar{r}(n, p) \leq \begin{cases} (n-1)p(p^{[(n-1)/(p-1)]+1} - 1), & \text{if } p \neq 2; \\ 4(n-1)(2^n - 1), & \text{if } p = 2. \end{cases}$$

PROOF. When $p > n$, the bound for $\bar{r}(n, p)$ in the corollary follows from Lemmas 6 and 12, and when $p \leq n$ the bound follows from Lemma 12, because solvability of the system

$$\sum_{i=1}^s \binom{x_i}{j} \equiv A_j \pmod{p^{\alpha_j}} \quad (j = 1, \dots, n), \text{ where } \alpha_j = \kappa + \delta + \left\lceil \frac{n-j}{p-1} \right\rceil,$$

for any integers $A_1, \dots, A_n$ can be shown by successively choosing for $j = n, n-1, \dots, 1$ at most $p^{\alpha_j} - 1$ terms x_i equal to j , in such a way that the j th congruence holds, i.e., it suffices to have the conditions

$$s \geq \sum_{j=1}^n (p^{\alpha_j} - 1) \quad (\leq (n-1)p^{\delta+1}(p^{[(n-1)/(p-1)]+1} - 1) - n).$$

In order to strengthen the estimate in the corollary, instead of the system of congruences

$$\sum_{i=1}^s \binom{x_i}{j} \equiv A_j \pmod{p^{\gamma + [(n-j)/(p-1)]}} \quad (j = 1, \dots, n), \quad (23)$$

where $0 \leq \gamma \leq \kappa + \delta$ and $A_1, \dots, A_n$ are integers, we consider another system of congruences, whose solvability ensures the solvability of (23) (and with the same values of the variables).

We set $\Phi_m(x) = 1 - f_{m(p-1)}(x - p + 1)$ ($m = 1, 2, \dots$), where $f_{m(p-1)}(x)$ is the polynomial in Lemma 7. Then the following congruences hold for integers x :

$$\Phi_m(x) \equiv \begin{cases} 1 \pmod{p^m}, & \text{if } x \equiv -1 \pmod{p}; \\ 0 \pmod{p^m}, & \text{if } x \not\equiv -1 \pmod{p}. \end{cases}$$

Let $j = m(p-1) - l$, where $0 \leq l \leq p-2$. Then $m = [(j-1)/(p-1)] + 1$. We set $F_j^*(x) = \Delta_1^{(l)} \Phi_m(x) = \sum_{\nu=0}^j a_{j,\nu} \binom{x}{\nu}$. Since (see [12], Chapter 1, §3) $\Delta_1^{(l)}(x) = \sum_{i=0}^l (-1)^{l-i} \binom{l}{i} f(x+i)$, it follows that we have the congruences:

$$F_j^*(x) \equiv \begin{cases} (-1)^{l-i} \binom{l}{i} \pmod{p^m}, & \text{if } x \equiv -1 - i \pmod{p}, 0 \leq i \leq l; \\ 0 \pmod{p^m} & \text{otherwise;} \end{cases} \quad (24)$$

in particular, $F_j^*(x) \equiv 0 \pmod{p^m}$, if $x \equiv 0 \pmod{p}$. We set $F_j(x) = F_j^*(x) - F_j^*(0)$. Then $F_j(x) = \sum_{\nu=1}^j a_{j,\nu} \binom{x}{\nu}$ and $F_j(x)$ satisfies the same congruences as $F_j^*(x)$. Since $a_{j,j} = a_{m(p-1), m(p-1)}$, and $-a_{m(p-1), m(p-1)}$ is the leading coefficient in the expansion of $f_{m(p-1)}(x)$ with respect to $\binom{x}{i}$ ($i = 1, \dots, m(p-1)$), it follows from Lemma 7 that $p^{m-1} \parallel a_{j,j}$. We set

$$\varphi_j(x) = p^{-m} \Delta_p F_j(x) = \sum_{\nu=0}^{j-1} b_{j,\nu} \binom{x}{\nu}. \quad (25)$$

From the congruences (24) it follows that $\varphi_j(x)$ is an integer-valued polynomial. By Lemma 1, the coefficients $b_{j,\nu}$ are integers, and since $b_{j,j-1} = p^{-m+1} a_{j,j}$, it follows that $b_{j,j-1} \not\equiv 0 \pmod{p}$. From (25) and Lemma 4 it follows that the $a_{j,\nu}$ satisfy

$$a_{j,\nu} \equiv 0 \pmod{p^{[(j-1)/(p-1)] - [(j-\nu)/(p-1)]}} \quad (1 \leq \nu \leq j). \quad (26)$$

We now consider the system of congruences

$$\sum_{i=1}^s F_j(x_i) \equiv B_j \pmod{p^{\gamma + [(n-1)/(p-1)]}} \quad (j = 1, \dots, n), \quad (27)$$

where

$$B_j = \sum_{v=1}^j a_{j,v} A_v, \quad (28)$$

which is connected with (23). Suppose that this system is satisfied for certain integers $x_1, \dots, x_s$. If we make successive linear transformations of (27), starting with $j = 1$, and use (26) and the relations

$$p^{[(j-1)/(p-1)]} \| a_{j,j}, \quad \left\lfloor \frac{n-1}{p-1} \right\rfloor \geq \left\lfloor \frac{n-j}{p-1} \right\rfloor + \left\lfloor \frac{j-1}{p-1} \right\rfloor, \\ \left(\gamma + \left\lfloor \frac{n-v}{p-1} \right\rfloor \right) + \left(\left\lfloor \frac{j-1}{p-1} \right\rfloor - \left\lfloor \frac{j-v}{p-1} \right\rfloor \right) \geq \gamma + \left\lfloor \frac{n-j}{p-1} \right\rfloor + \left\lfloor \frac{j-1}{p-1} \right\rfloor,$$

we arrive at (23). We have thereby reduced (23) to the study of (27).

LEMMA 13. Suppose that $p \leq n$; $\alpha_1, \dots, \alpha_q$ are integers, and

$$s \geq \sum_{n-p+1 < j \leq n} (p^{[(j-1)/(p-1)]} - 1).$$

Then the system of congruences

$$\sum_{i=1}^q F_j(\alpha_i) + \sum_{i=1}^s F_j(x_i) \equiv B_j \pmod{p^{[(j-1)/(p-1)]+1}} \quad (j = 1, \dots, n) \quad (29)$$

is solvable in integers $x_1, \dots, x_s$.

PROOF. It follows from (26) and (28) that

$$B_j \equiv B_{j-p+1} \pmod{p^{[(j-1)/(p-1)]}}, \quad (30)$$

if $j \geq p$. Since by the construction of the polynomials $f_n(x)$ we have $f_{m(p-1)}(x) \equiv f_{(m-1)(p-1)}(x) \pmod{p^{m-1}}$ for integers x , it then follows that also

$$F_j(x) \equiv F_{j-p+1}(x) \pmod{p^{[(j-1)/(p-1)]}} \quad (j \geq p) \quad (31)$$

for integers x . The congruences (30) and (31) show that in (29) it suffices to restrict oneself to consideration of the following $p-1$ congruences:

$$\sum_{i=1}^q F_j(\alpha_i) + \sum_{i=1}^s F_j(x_i) \equiv B_j \pmod{p^{[(j-1)/(p-1)]+1}}, \quad n-p+2 \leq j \leq n. \quad (32)$$

Let $n = \beta(p-1) - \alpha$, where $0 \leq \alpha \leq p-2$, $\beta = [(n-1)/(p-1)] + 1$. If $k_0, k_1, \dots, k_{p-1}$ is, respectively, the number of variables which run through values congruent to $0, 1, \dots, p-1$ modulo p , then, by (24), system (32) is equivalent to

$$\left\{ \begin{array}{l} \sum_{i=1}^q F_j(\alpha_i) + \sum_{i=0}^l k_{p-1-i} (-1)^{l-i} \binom{l}{i} \equiv B_j \pmod{p^\beta}, \\ j = n + \alpha - l, \alpha \leq l \leq p-2, \\ \sum_{i=1}^q F_j(\alpha_i) + \sum_{i=0}^l k_{p-1-i} (-1)^{l-i} \binom{l}{i} \equiv B_j \pmod{p^{\beta-1}}, \\ j = n + \alpha - (p-1+l), 0 \leq l \leq \alpha-1. \end{array} \right.$$

This is a linear system with respect to $k_1, \dots, k_{p-1}$, it has a triangular form, and hence it is solvable under the conditions $0 \leq k_{p-1}, \dots, k_{p-\alpha} \leq p^{\beta-1} - 1$ and $0 \leq k_{p-\alpha+1}, \dots, k_1 \leq p^\beta - 1$. Since $k_1 + \dots + k_{p-1} \leq s$, the lemma is proved.

LEMMA 14. Let $p \leq n$, $\kappa = [\ln(n-1)/\ln p] + 1$, and $\delta = 1 - 2\{p/2\}$. Then

$$\bar{r}(n, p) \leq pr(l, p) + \sum_{n-p+1 < j \leq n} (p^{[(j-1)/(p-1)]+1} - 1) + n, \quad (33)$$

where l is the smallest natural number such that the following inequality holds for $v > l$ and for all j in the interval $[v, n]$:

$$\max\left(0, v - 1 - \left\lfloor \frac{j-v}{p-1} \right\rfloor\right) \geq \left\lfloor \frac{n-1}{p-1} \right\rfloor - \left\lfloor \frac{j-1}{p-1} \right\rfloor + \kappa + \delta - 1. \quad (34)$$

PROOF. Suppose that l is the natural number defined in the lemma, $R = r(l, p)$, $\gamma(j) = [(j-1)/(p-1)] + 1$ ($j = 1, \dots, n$); $A_1, \dots, A_n$ are integers, and $B_1, \dots, B_n$ are the integers given by (28). We set $d = \sum_{n-p+1 < j \leq n} (p^{\gamma(j)} - 1)$, and, in accordance with Lemma 13, we determine integers $x_1, \dots, x_d$ such that the following system of congruences is satisfied:

$$R \sum_{\alpha=0}^{p-1} F_j(\alpha) + \sum_{i=1}^d F_j(x_i) \equiv B_j \pmod{p^{\gamma(j)}} \quad (j = 1, \dots, n). \quad (35)$$

We consider the system of congruences ($j = 1, \dots, n$)

$$\sum_{\alpha=0}^{p-1} \sum_{\beta=1}^R F_j(\alpha + py_{\alpha,\beta}) + \sum_{i=1}^d F_j(x_i) \equiv B_j \pmod{p^{\kappa+\delta+[(n-1)/(p-1)]}} \quad (36)$$

with integer variables $y_{\alpha,\beta}$. As noted earlier in this section, if we prove that this system is solvable, then, by Lemma 12, we have the required estimate $\bar{r}(n, p) \leq pR + d + n$. By (35), system (36) is equivalent to the system ($j = 1, \dots, n$)

$$\sum_{\alpha=0}^{p-1} \sum_{\beta=1}^R p^{-\gamma(j)} \Delta_{py_{\alpha,\beta}} F_j(\alpha) \equiv c_j \pmod{p^{\kappa+\delta+[(n-1)/(p-1)]-\gamma(j)}}$$

and it is further equivalent, by (25) and Corollary 2 of Lemma 2, to the system ($j = 1, \dots, n$)

$$\sum_{\alpha=0}^{p-1} \sum_{\beta=1}^R \left(\sum_{\nu=1}^j \Delta_p^{(\nu-1)} \varphi_j(\alpha) \binom{y_{\alpha,\beta}}{\nu} \right) \equiv c_j \pmod{p^{\kappa+\delta+[(n-1)/(p-1)]-\gamma(j)}} \quad (37)$$

with certain integers $c_1, \dots, c_n$. The polynomial $\varphi_j(x)$ has degree $j-1$. According to Lemma 8, $\Delta_p^{(\nu-1)} \varphi_j(\alpha)$ is divisible by p to the power

$$\max\left(0, \nu - 1 - \left\lfloor \frac{j-\nu}{p-1} \right\rfloor\right),$$

which is $\geq \kappa + \delta + [(n-1)/(p-1)] - \gamma(j)$ for $\nu > l$. Consequently, (37) is equivalent to the system ($j = 1, \dots, n$)

$$\sum_{\alpha=0}^{p-1} \sum_{\beta=1}^R \left(\sum_{\nu=1}^l \Delta_p^{(\nu-1)} \varphi_j(\alpha) \binom{y_{\alpha,\beta}}{\nu} \right) \equiv c_j \pmod{p^{\kappa+\delta+[(n-1)/(p-1)]-\gamma(j)}}. \quad (38)$$

To prove that the system of congruences (38) is solvable it is sufficient to prove that the system of equations

$$\sum_{\alpha=0}^{p-1} \sum_{\beta=1}^R \left(\sum_{\nu=1}^l \Delta_p^{(\nu-1)} \varphi_j(\alpha) \binom{y_{\alpha,\beta}}{\nu} \right) = c_j \quad (j = 1, \dots, n) \quad (39)$$

is solvable in p -adic integers $y_{\alpha,\beta}$. Since

$$\varphi_j(x) = \sum_{i=0}^{j-1} b_{j,i} \binom{x}{i}, \quad b_{j,j-1} \not\equiv 0 \pmod{p},$$

it follows that (39) is equivalent to the system

$$\sum_{\alpha=0}^{p-1} \sum_{\beta=1}^R \left(\sum_{\nu=1}^l \Delta_p^{(\nu-1)} \binom{\alpha}{j-1} \binom{y_{\alpha,\beta}}{\nu} \right) = M_j \quad (j = 1, \dots, n) \quad (40)$$

with certain p -adic integers M_j .

We set $\lambda_{\nu,j} = \Delta_p^{(\nu-1)} \binom{0}{j-1}$ ($\nu, j = 1, 2, \dots$) and $\lambda_{\nu,j} = 0$ for $j \leq 0$. Since, by Lemma 2,

$$\binom{x+\alpha}{j-1} = \sum_{i=0}^{\min(\alpha, j-1)} \binom{x}{j-1-i} \binom{\alpha}{i}, \quad \alpha \geq 0,$$

it follows that

$$\Delta_p^{(\nu-1)} \binom{\alpha}{j-1} = \sum_{i=0}^{\alpha} \lambda_{\nu,j-i} \binom{\alpha}{i}, \quad \alpha \geq 0. \quad (41)$$

Using Lemma 2, we also find that ($\nu > 1$)

$$\begin{aligned} \lambda_{\nu,j} &= \sum_{\substack{1 \leq i_1, \dots, i_{\nu-1} \leq p \\ i_1 + \dots + i_{\nu-1} \leq j-1}} \binom{0}{j-1-i_1-\dots-i_{\nu-1}} \binom{p}{i_1} \cdots \binom{p}{i_{\nu-1}} \\ &\equiv \begin{cases} 1 \pmod{p}, & \text{if } (\nu-1)p = j-1; \\ 0 \pmod{p} & \text{otherwise.} \end{cases} \end{aligned} \quad (42)$$

We introduce the notation

$$Y_{\nu,\alpha} = \sum_{\beta=1}^R \binom{y_{\alpha,\beta}}{\nu} \quad (\alpha = 0, \dots, p-1; \nu = 1, \dots, l), \quad (43)$$

$$X_{\nu,i} = \sum_{\alpha \leq p-1} \binom{\alpha}{i} Y_{\nu,\alpha} \quad (i = 0, \dots, p-1; \nu = 1, \dots, l). \quad (44)$$

Then, taking (41) into account, we can write (40) in the form

$$\sum_{\nu=1}^l \sum_{i=0}^{p-1} \lambda_{\nu,j-1} X_{\nu,i} = M_j \quad (j = 1, \dots, n). \quad (45)$$

From the definition of l it follows that $l \geq [(n+p-1)/p]$. In (45) we choose $X_{\nu,i} = 0$ if $\nu > [(n+i)/p]$. Then, by (42), system (45) in the remaining n variables $X_{\nu,i}$ has determinant $\not\equiv 0 \pmod{p}$. Consequently, (45) is solvable in p -adic integers $X_{\nu,i}$. From (44) we now find the p -adic integers $Y_{\nu,\alpha}$, and then, keeping in mind the property of $r(l, p) = R$ at the beginning of the section, from (43) we find the p -adic integers $y_{\alpha,\beta}$. The lemma is proved.

PROPOSITION 1. *The following estimate holds for $p \geq 3$:*

$$\bar{r}(n, p) \leq p^{(n-1)/(p-1)+2}.$$

PROOF. For $n < p$ we have, by the corollary to Lemma 12,

$$\bar{r}(n, p) \leq pn < p^{(n-1)/(p-1)+2}.$$

Suppose that $n \geq p$ and the estimate of Proposition 1 holds for all natural numbers $m < n$. We estimate $\bar{r}(n, p)$ by applying Lemma 14. We set

$$v_0 = \left\lceil \frac{p-1}{p} \left(\left\lceil \frac{n-1}{p-1} \right\rceil + \left\lceil \frac{\ln(n-1)}{\ln p} \right\rceil \right) \right\rceil + 2.$$

Then for $\nu = v_0$ and $p \geq 3$ inequality (34) holds for all integers j . Consequently, in (33) we can take

$$l \leq v_0 - 1 = \left\lceil \frac{p-1}{p} \left(\left\lceil \frac{n-1}{p-1} \right\rceil + \left\lceil \frac{\ln(n-1)}{\ln p} \right\rceil \right) \right\rceil + 1 < n.$$

We represent $n-1$ in the form $n-1 = t(p-1) + \lambda$, where t and λ are integers, $t \geq 1$, and $0 \leq \lambda \leq p-2$. Since $\lambda + 2 \leq (p+1)p^{\lambda/(p-1)}/2$, it follows that

$$\begin{aligned} \sum_{i=1}^{p-1} p^{[(n-i)/(p-1)]+1} &= (\lambda+1)p^{t+1} + (p-2-\lambda)p^t = (\lambda+2)(p-1)p^t \\ &\leq \frac{p^2-1}{2} p^{(n-1)/(p-1)}. \end{aligned} \quad (46)$$

If $n = p$, then $l = 1$, $\bar{r}(l, p) = 1$, and, by (33),

$$\bar{r}(p, p) \leq 2p + 2(p-1)p = 2p^2 < p^{(n-1)/(p-1)+2}. \quad (47)$$

Suppose that $n > p$ and $t = 1$. Then $l = 2$, $\bar{r}(l, p) \leq 2p$, and, since $p < n \leq p^{(n-1)/(p-1)}$, it follows from (33) and (46) that

$$\begin{aligned} \bar{r}(n, p) &< 2p^2 + \frac{p^2+1}{2} p^{(n-1)/(p-1)} \\ &\leq \frac{p^2-1}{2} p^{p/(p-1)} + \frac{p^2+1}{2} p^{(n-1)/(p-1)} \leq p^{(n-1)/(p-1)+2}. \end{aligned}$$

Now suppose that $t \geq 2$. If $n \leq (p-1)^2$, then $l \leq p-1$, $\bar{r}(l, p) \leq p(p-1)$, and from (33) and (46) it follows that

$$\bar{r}(n, p) < p^2(p-1) + \frac{p^2+1}{2} p^{(n-1)/(p-1)} < p^{(n-1)/(p-1)+2}.$$

If $(p-1)^2 < n \leq p(p-1)+1$, then $l \leq p$, $\bar{r}(l, p) \leq \bar{r}(p, p) \leq 2p^2$ by (47), and, by (33) and (46), $\bar{r}(n, p) < 2p^3 + (p^2+1)p^{(n-1)/(p-1)}/2 \leq p^{(n-1)/(p-1)+2}$. Finally, suppose that $n \geq p(p-1)+2$. Then $n \leq \frac{1}{2}p^{(n-1)/(p-1)}$. If we apply this inequality, along with (33), (46), and the induction assumption, we obtain

$$\bar{r}(n, p) \leq p\bar{r}(l, p) + \frac{1}{2}p^{(n-1)/(p-1)+2} \leq p^{(l-1)/(p-1)+3} + \frac{1}{2}p^{(n-1)/(p-1)+2}.$$

In order to get the required inequality, it is now sufficient to show that

$$\frac{l-1}{p-1} \leq \frac{n-1}{p-1} - 1 - \ln 2. \quad (48)$$

Since

$$\frac{l-1}{p-1} \leq \frac{n-1}{p(p-1)} + \frac{1}{p} \left\lceil \frac{\ln(n-1)}{\ln p} \right\rceil,$$

(48) will follow from the inequality

$$n - 1 \geq [\ln(n - 1)/\ln p] + p(1 + \ln 2). \quad (49)$$

We denote $k = [\ln(n - 1)/\ln p]$. If $k = 1$, then (49) follows immediately from the condition $n - 1 \geq p(p - 1) + 1$. If $k \geq 2$, then $n - 1 \geq p^k \geq p(1 + (p - 1)(k - 1)) > k + 2p > k + p(1 + \ln 2)$, and (49) holds. Thus, in all cases,

$$\bar{r}(n, p) \leq p^{(n-1)/(p-1)+2}.$$

The proposition is proved.

LEMMA 15. Suppose that $n \geq 2$, and the following bound holds for all p : $\bar{r}(n, p) \leq R(n)$, where $R(n) \geq n^2 + 2$. Then when $s \geq R(n)$ the singular series σ in (2) is bounded from below by a positive constant depending only on n , for any natural numbers $N_1, \dots, N_n$ which satisfy the arithmetic necessary conditions for system (1) to be solvable.

PROOF. Let $A_1, \dots, A_n$ be the integers in terms of which $N_1, \dots, N_n$ are determined by (14). We set $\kappa = [\ln(n - 1)/\ln p] + 1$ and $\delta = 1 - 2\{p/2\}$. By the definition of $\bar{r}(n, p)$, for any integers z_i ($i = 1, \dots, s - R(n)$) there exist integers y_α ($\alpha = 1, \dots, R(n) - n$) and a lifting tuple $(\lambda_1, \dots, \lambda_n)$ modulo p such that the following system of congruences holds:

$$\sum_{i=1}^{s-R(n)} \binom{z_i}{j} + \sum_{\alpha=1}^{R(n)-n} \binom{y_\alpha}{j} + \sum_{i=1}^n \binom{\lambda_i}{j} \equiv A_j \pmod{p^{\beta_j}} \quad (j = 1, \dots, n),$$

where $\beta_j = [(n - j)/(p - 1)] + \kappa + \delta$. According to the corollary of Lemma 3, the number p^t , where $t = \beta_1 + [\ln n / \ln p]$, is a period modulo p^{β_1} for all integer-valued polynomials of degree at most n . Thus, if we apply Lemma 12, we find that the system of congruences ($j = 1, \dots, n$)

$$\sum_{i=1}^{s-R(n)} \binom{z_i}{j} + \sum_{\alpha=1}^{R(n)-n} \binom{y_\alpha + p^t u_\alpha}{j} + \sum_{i=1}^n \binom{x_i}{j} \equiv A_j \pmod{p^{\beta_j+k}}$$

is solvable in integers $x_1, \dots, x_n$ for any integers z_i and u_α and any integer $k \geq 0$. In view of (14), this implies that the system of congruences ($j = 1, \dots, n$)

$$\sum_{i=1}^{s-R(n)} z_i^j + \sum_{\alpha=1}^{R(n)-n} (y_\alpha + p^t u_\alpha)^j + \sum_{i=1}^n x_i^j \equiv N_j \pmod{p^{\beta_n+k}}$$

is solvable for any integers z_i and u_α and any $k \geq 0$.

We let $W(p^m, s)$ denote the number of solutions of the system of congruences $\sum_{i=1}^s x_i^j \equiv N_j \pmod{p^m}$ ($j = 1, \dots, n$). Then for $m \geq t$ we have

$$p^{m(n-s)} W(p^m, s) \geq p^{m(s-R(n)) + (m-t)(R(n)-n) + m(n-s)} = p^{-t(R(n)-n)}. \quad (50)$$

We now estimate $W(p^m, s)$ in a second way. We write $W(p^m, s)$ in terms of trigonometric sums:

$$\begin{aligned} W(p^m, s) &= p^{m(s-n)} + p^{-mn} \sum_{r=1}^m \sum_{\substack{1 \leq a_1, \dots, a_n \leq p^r \\ (a_1, \dots, a_n, p) = 1}} p^{(m-r)s} \\ &\quad \times \left(\sum_{x=1}^{p^r} e^{2\pi i(a_1 x + \dots + a_n x^n)/p^r} \right)^s e^{-2\pi i(a_1 N_1 + \dots + a_n N_n)/p^r}. \end{aligned}$$

Applying the well-known estimate (see [20])

$$\left| \sum_{x=1}^{p^k} e^{2\pi i(a_1 x + \dots + a_n x^n)/p^k} \right| \leq c(n) p^{k(1-1/l)},$$

where $p > n$, $(a_1, \dots, a_n, p) = 1$, and $l = l(n, k) = \min(n, \max(k, 2))$, we conclude that for $p > p_0(n)$ and $s \geq n^2 + 2$

$$\begin{aligned} p^{m(n-s)} W(p^m, s) &\geq 1 - p^{-ms} \sum_{r=1}^m p^{nr+(m-r)s} (c(n))^s p^{r(1-1/l(n,r))s} \\ &\geq 1 - (c(n))^s (p^{-s/2+n} + 3p^{-s+n^2}) \geq 1 - p^{-3/2}. \end{aligned} \quad (51)$$

If we use (50) for $p \leq p_0(n)$ and (51) for $p > p_0(n)$, by Lemma 9 we obtain the lower bound for σ :

$$\sigma \geq \prod_{p \leq p_0(n)} p^{-t(R(n)-n)} \prod_{p > p_0(n)} (1 - p^{-3/2}) = c_0(n) > 0.$$

The lemma is proved.

§3. Upper bound for $r(n)$

For $p = 2$ the recursive inequality (33), combined with the estimate in the corollary to Lemma 12 and the lower bound $r(n) \geq 2^n - 1$, gives the asymptotic formula $\bar{r}(n, 2) = 2^n + O(\sqrt{n} 2^{n/2})$. In order to obtain a more precise upper bound for $\bar{r}(n, 2)$, and hence for $r(n)$, in this section we shall consider a system of equations in the ring of 2-adic integers which has a somewhat more general form than (13).

Suppose that $m_1, \dots, m_q$ are natural numbers which satisfy the conditions $m_1 + \dots + m_q = n$, $|m_i - m_j| \leq 1$ ($i, j = 1, \dots, q$), the t_i ($i = 1, \dots, [q/2]$) are a sequence of natural numbers with $2i - 1 \leq t_i \leq 2i$, and $m_0 = \max(m_1, \dots, m_q)$. We consider the system of equations

$$\begin{aligned} \sum_{t=1}^q \left(\sum_{i=1}^{s_i} \left(k_{\lambda,t} \binom{x_{i,t}}{j} - 2\sigma_{\lambda,j}^{(t)} \binom{x_{i,t}}{j+1} \right) + \varphi_{\lambda,j}^{(t)}(x_{i,t}) \right) - \sum_{i=1}^{s_{\beta(t)}} 2k_{\lambda,t} \sigma_{\lambda,j}^{(t)} \binom{x_{i,\beta(t)}}{j+1} &= A_{\lambda,j} \\ (j = 1, \dots, m_\lambda; \lambda = 1, \dots, q), \end{aligned} \quad (52)$$

where the $A_{\lambda,j}$ are 2-adic integers, the $k_{\lambda,t}$ are 2-adic integers which satisfy the congruences $k_{\lambda,\lambda} \equiv 1 \pmod{2}$ and $k_{\lambda,t} \equiv 0 \pmod{2}$ for $t > \lambda$, the polynomials $\varphi_{\lambda,j}^{(t)}(x)$ have the form

$$\varphi_{\lambda,j}^{(t)}(x) = \sum_{\nu=1}^M 2^{\max(2m_\lambda+1-\nu, \nu)+1-j} c_{\nu,\lambda,j}^{(t)} \binom{x}{\nu},$$

where the $c_{\nu,\lambda,j}^{(t)}$ are 2-adic integers, $M \geq n$,

$$\beta(t) = \begin{cases} t+1, & \text{if } m_0 = 1; \\ 4[(t+1)/2] - t - 1, & \text{if } m_0 > 1, \end{cases}$$

and the $\sigma_{\lambda,j}^{(t)}$ take the value 0 or 1; the value 1 is taken only if m_λ is odd, $t < q$, $\lambda < q$, $j = m_\lambda$, $m_\lambda = m_{\lambda+1}$, $k_{\lambda+1,\lambda} \equiv 1 \pmod{2}$, and if $m_0 > 1$, then we also require that q be even, λ be odd, t and λ be elements of the sequence $t_1, t_2, \dots$, and

$$k_{\lambda,t_i} \equiv k_{\lambda+1,2i-1} \equiv k_{\lambda+1,2i} \pmod{2}, \quad k_{\lambda,\beta(t_i)} \equiv 0 \pmod{2},$$

$$\sigma_{\lambda,m_\lambda}^{(t_i)} = 1 \quad \left(i = 1, \dots, \frac{q}{2} \right).$$

We let $R(m_t)$ ($t = 1, \dots, q$) denote the smallest number (if such a number exists) of variables $x_{i,t}$ such that system (52) is solvable in 2-adic integers $x_{i,t}$ which contain a rank- m_t lifting tuple (mod 2) for each $t = 1, \dots, q$, for any $A_{\lambda,j}$, $k_{\lambda,t}$, $\varphi_{\lambda,j}^{(t)}$, $\sigma_{\lambda,j}^{(t)}$, and sequence $t_1, t_2, \dots$ which satisfy our conditions. Since (52) does not change its form if some of the variables $x_{i,t}$ are given concrete values for each t , it follows that the q -tuple $R(m_1), \dots, R(m_q)$ exists whenever (52) is solvable for some particular $s_1, \dots, s_q$ for all $A_{\lambda,j}$, $k_{\lambda,t}$, $\varphi_{\lambda,j}^{(t)}$, $\sigma_{\lambda,j}^{(t)}$, and t_i which satisfy our conditions.

LEMMA 16. Suppose that $n \geq 2$, $n_0 = [n/2]$, $n_1 = [(n+1)/2]$, and $a_1, \dots, a_{n_0}$ and $b_1, \dots, b_{n_1}$ are lifting tuples (mod 2) of rank n_0 and n_1 , respectively. Then either of the sets of numbers

$$1 + 2a_1, \dots, 1 + 2a_{n_0}, 2b_1, \dots, 2b_{n_1} \quad (53)$$

or

$$2a_1, \dots, 2a_{n_0}, 1 + 2b_1, \dots, 1 + 2b_{n_1} \quad (54)$$

is a rank- n lifting tuple (mod 2).

PROOF. We have $n = n_0 + n_1$. Let $c_1, \dots, c_n$ be either of the tuples (53) or (54). Then the definition of rank- n_0 and rank- n_1 lifting tuples (mod 2) implies that

$$\begin{aligned} \det \left(\binom{c_i}{j} \right)_{\substack{i=1, \dots, n \\ j=0, \dots, n-1}} &= \prod_{1 \leq l < k \leq n} \frac{c_k - c_l}{k - l} \\ &\equiv \left(\prod_{1 \leq l < k \leq n_0} \frac{a_k - a_l}{k - l} \right) \left(\prod_{1 \leq l < k \leq n_1} \frac{b_k - b_l}{k - l} \right) \not\equiv 0 \pmod{2} \end{aligned}$$

and hence $c_1, \dots, c_n$ is a rank- n lifting tuple (mod 2). The lemma is proved.

PROPOSITION 2. For any natural numbers $m_1, \dots, m_q$ satisfying the conditions $m_1 + \dots + m_q = n$ and $|m_i - m_j| \leq 1$ ($i, j = 1, \dots, q$) the q -tuple $R(m_1), \dots, R(m_q)$ exists and satisfies the bound $R(m_t) \leq H(m_t)$ ($t = 1, \dots, q$), where

$$H(m) = \sum_{0 \leq k \leq [\ln n / \ln 2]} 2^k (2^{\lfloor m/2^k \rfloor} - 1).$$

PROOF. We use induction on m_0 . Suppose that $m_0 = 1$. Then $m_1 = \dots = m_q = 1$, $q = n$. Since $2^\alpha \parallel \nu!$, it follows that $\alpha < \nu - 1$. Hence, the polynomials $\frac{1}{2}\varphi_{\lambda,1}^{(t)}(x)$ have 2-adic integer coefficients. We set $s_1 = \dots = s_q = 1$. Since $k_{\lambda,t} \equiv 0 \pmod{2}$ for $t > \lambda$, it follows that (52) takes the form

$$\sum_{t=1}^{\lambda} k_{\lambda,t} \left(x_t - 2\sigma_{\lambda,1}^{(t)} \left(\binom{x_t}{2} + \binom{x_{t+1}}{2} \right) \right) + 2\psi_{\lambda}(x_1, \dots, x_q) = A_{\lambda,1} \quad (\lambda = 1, \dots, q), \quad (55)$$

where the $\psi_{\lambda}(x_1, \dots, x_q)$ are polynomials with 2-adic integer coefficients. If $\lambda < q$ and $\sigma_{\lambda,1}^{(\lambda)} = 1$, then, because of the congruences $k_{\lambda,\lambda} \equiv 1 \pmod{2}$ and $k_{\lambda+1,\lambda} \equiv k_{\lambda+1,\lambda+1} \equiv 1 \pmod{2}$, the equations in (55) corresponding to λ and $\lambda + 1$ can be represented in the following form, respectively:

$$\begin{aligned} -x_{\lambda}^2 - x_{\lambda+1}^2 + x_{\lambda+1} + \theta_{\lambda}(x_1, \dots, x_{\lambda-1}) + 2\psi_{\lambda}(x_1, \dots, x_q) &= 0, \\ x_{\lambda} + x_{\lambda+1} + \theta_{\lambda+1}(x_1, \dots, x_{\lambda-1}) + 2\psi_{\lambda+1}(x_1, \dots, x_q) &= 0, \end{aligned}$$

where θ_λ and $\theta_{\lambda+1}$ are polynomials with 2-adic integer coefficients. After substituting $x_{\lambda+1}$ from the second equation into the first, we obtain the equivalent equations

$$\begin{aligned} x_\lambda + \theta_\lambda^*(x_1, \dots, x_{\lambda-1}) + 2\psi_\lambda^*(x_1, \dots, x_q) &= 0, \\ x_{\lambda+1} + \theta_{\lambda+1}^*(x_1, \dots, x_\lambda) + 2\psi_{\lambda+1}^*(x_1, \dots, x_q) &= 0, \end{aligned} \quad (56)$$

where θ_λ^* , $\theta_{\lambda+1}^*$, ψ_λ^* , and $\psi_{\lambda+1}^*$ are polynomials with 2-adic integer coefficients. If $\sigma_{\lambda,1}^{(\lambda)} = 0$, then the equation in (55) corresponding to λ already has the form (56). It follows from Lemma 5 that a system of equations of the form (56) ($\lambda = 1, \dots, q$) is solvable in 2-adic integers $x_1, \dots, x_q$. Since $H(1) = 1$, and we can take any 2-adic integer as our rank-1 lifting tuple (mod 2), we have thereby proved that the proposition holds for $m_0 = 1$.

Now suppose that $m_0 > 1$, and that the q -tuple $R(n_1), \dots, R(n_q)$ with bound $R(n_i) \leq H(n_i)$ ($i = 1, \dots, q$) exists for all natural numbers q and $n_1, \dots, n_q$ which satisfy $n_1 + \dots + n_q = n$ and $|n_i - n_j| \leq 1$ ($i, j = 1, \dots, q$) with $\max(n_1, \dots, n_q) < m_0$. We consider system (52), where we set $s_t = 2^{m_t} - 1 + 2r_t$, $r_t = H(\lfloor m_t/2 \rfloor)$, and $H(0) = 0$ ($t = 1, \dots, q$). We multiply the equations corresponding to j by $(-2)^{j-1}$, and for each λ and $l \leq m_\lambda$ we add the resulting equations as j goes from 1 to l . We then obtain the following system of equations which is equivalent to (52):

$$\begin{aligned} \sum_{t=1}^q \left(\sum_{i=1}^{s_t} \left(k_{\lambda,t} \left(f_l(x_{i,t}) + (-2)^{m_\lambda} \sigma_{\lambda,l}^{(t)} \binom{x_{i,t}}{l+1} \right) + F_{\lambda,l}^{(t)}(x_{i,t}) \right) \right. \\ \left. + \sum_{i=1}^{s_{\beta(t)}} (-2)^{m_\lambda} k_{\lambda,t} \sigma_{\lambda,l}^{(t)} \binom{x_{i,\beta(t)}}{l+1} \right) = B_{\lambda,l} \\ (l = 1, \dots, m_\lambda; \lambda = 1, \dots, q), \end{aligned} \quad (57)$$

where

$$f_l(x) = \sum_{j=1}^l (-2)^{j-1} \binom{x}{j}, \quad B_{\lambda,l} = \sum_{j=1}^l (-2)^{j-1} A_{\lambda,j}, \quad (58)$$

$$F_{\lambda,l}^{(t)}(x) = \sum_{\nu=1}^M 2^{\max(2m_\lambda+1-\nu, \nu)} b_{\nu,\lambda,l}^{(t)} \binom{x}{\nu}, \quad b_{\nu,\lambda,l}^{(t)} = \sum_{j=1}^l (-1)^{j-1} c_{\nu,\lambda,j}^{(t)}. \quad (59)$$

We represent the variables $x_{i,t}$ in the form $z_{i,t} + 2u_{i,t}$. Setting $z_{i,t} = 0$ for $2^{m_t} \leq i \leq 2^{m_t} - 1 + r_t$ and $z_{i,t} = 1$ for $2^{m_t} + r_t \leq i \leq s_t$, and noting that $f_l(0) = 0$ and $f_l(1) = 1$, we can apply Lemma 10 and choose the remaining $z_{i,t}$ ($i = 1, \dots, 2^{m_t} - 1$) to be 0 or 1 in such a way that the following congruences hold ($l = m_\lambda; \lambda = 1, \dots, q$):

$$\sum_{i=1}^q \left(k_{\lambda,t} r_t + k_{\lambda,t} \sum_{i=1}^{2^{m_t}-1} f_l(z_{i,t}) \right) \equiv B_{\lambda,l} \pmod{2^l}. \quad (60)$$

Since, by Lemma 7, for integers x we have

$$f_l(x) \equiv \begin{cases} 0 \pmod{2^l}, & \text{if } x \equiv 0 \pmod{2}; \\ 1 \pmod{2^l}, & \text{if } x \equiv 1 \pmod{2}, \end{cases}$$

it follows from (60), (58), and (59) that for our chosen $z_{i,t}$ and for any 2-adic integers $u_{i,t}$ we can pass to congruences modulo 2^l in the equations in (57) corresponding to l . In order to know that (57) is solvable for the chosen $z_{i,t}$ it is now sufficient to prove that the

system

$$\sum_{t=1}^q \left(\sum_{i=1}^{s_t} \left(k_{\lambda,t} \left(2^{-l} \Delta_{2u_{i,t}} \left(f_l(z_{i,t}) + (-2)^{m_\lambda} \sigma_{\lambda,l}^{(t)} \left(\frac{z_{i,t}}{l+1} \right) \right) \right) + 2^{-l} \Delta_{2u_{i,t}} F_{\lambda,l}^{(t)}(z_{i,t}) \right) \right. \\ \left. + \sum_{i=1}^{s_{\beta(t)}} (-2)^{m_\lambda} k_{\lambda,t} \sigma_{\lambda,l}^{(t)} 2^{-l} \Delta_{2u_{i,\beta(t)}} \left(\frac{z_{i,\beta(t)}}{l+1} \right) \right) = C_{\lambda,l} \\ (l = 1, \dots, m_\lambda; \lambda = 1, \dots, q)$$

is solvable for some 2-adic integers $C_{\lambda,l}$.

For each t the numbers $z_{i,t}$ ($i = 1, \dots, 2^{m_t} - 1$) include at least 2^{m_t-1} which are equal, and for m a natural number we have

$$H([(m+1)/2]) - H([m/2]) \leq 2^{m-1};$$

hence, if we leave alone the $H([(m_t+1)/2]) - H([m_t/2])$ variables $u_{i,t}$ with $i \in [1, 2^{m_t} - 1]$ which correspond to the same value of $z_{i,t}$ and if we set the other variables equal to zero, then after changing notation from $z_{i,t} + 2u_{i,t}$ to $\varepsilon + 2u_{i,t}^{(\varepsilon)}$ and reindexing we obtain the system ($l = 1, \dots, m_\lambda; \lambda = 1, \dots, q$)

$$\sum_{\varepsilon=0}^1 \sum_{t=1}^q \left(\sum_{i=1}^{s_{\varepsilon,t}} \left(k_{\lambda,t} \left(2^{-l} \Delta_{2u_{i,t}^{(\varepsilon)}} \left(f_l(\varepsilon) + (-2)^{m_\lambda} \sigma_{\lambda,l}^{(t)} \left(\frac{\varepsilon}{l+1} \right) \right) \right) + 2^{-l} \Delta_{2u_{i,t}^{(\varepsilon)}} F_{\lambda,l}^{(t)}(\varepsilon) \right) \right. \\ \left. + \sum_{i=1}^{s_{\varepsilon,\beta(t)}} (-1)^{m_\lambda} k_{\lambda,t} \sigma_{\lambda,l}^{(t)} \Delta_{2u_{i,\beta(t)}^{(\varepsilon)}} \left(\frac{\varepsilon}{l+1} \right) \right) = C_{\lambda,l}, \quad (61)$$

where one of the numbers $s_{0,t}, s_{1,t}$ is $H([(m_t+1)/2])$ and the other is $H([m_t/2])$.

We note that a proof that (61) is solvable under the condition that for each t there is a lifting tuple (mod 2) among the $u_{i,t}^{(\varepsilon)}$ ($i = 1, \dots, s_{\varepsilon,t}$) which depends on ε and has rank $[(m_t+1)/2]$ or $[m_t/2]$, implies, by Lemma 16, that system (57) is solvable with rank- m_t lifting tuple (mod 2) among the $x_{i,t}$ for each t , and, since the $A_{\lambda,j}$, etc. are arbitrary, we obtain the required estimate $R(m_t) \leq 2^{m_t} - 1 + 2r_t = H(m_t)$ ($t = 1, \dots, q$).

We now transform system (61). We have

$$f_m(x) + (-2)^m \binom{x}{m+1} = f_{m+1}(x).$$

By the remark after Lemma 7, $\Delta_2 f_l(x) = (-1)^{l-1} 2^l \binom{x}{l-1}$ and so, denoting $\sigma = \sigma_{\lambda,l}^{(t)}$ for brevity, from Corollary 2 of Lemma 2 we obtain

$$2^{-l} \Delta_{2u} \left(f_l(\varepsilon) + (-2)^{m_\lambda} \sigma \binom{\varepsilon}{l+1} \right) = \sum_{\mu=1}^n (-1)^{l-1+\sigma} 2^{\sigma} \Delta_2^{(\mu-1)} \binom{\varepsilon}{l-1+\sigma} \binom{u}{\mu}, \quad (62)$$

$$2^{-l} \Delta_{2u} F_{\lambda,l}^{(t)}(\varepsilon) = \sum_{\mu=1}^M \sum_{\nu=1}^M 2^{\max(2m_\lambda+1-\nu, \nu)-l} b_{\nu,\lambda,l}^{(t)} \Delta_2^{(\mu)} \binom{\varepsilon}{\nu} \binom{u}{\mu}, \quad (63)$$

$$\Delta_{2u} \binom{\varepsilon}{l+1} = \sum_{\mu=1}^n \Delta_2^{(\mu)} \binom{\varepsilon}{l+1} \binom{u}{\mu}. \quad (64)$$

We set

$$\gamma_{\mu,j} = \begin{cases} \Delta_2^{(\mu-1)} \binom{0}{j-1}, & \text{if } j \geq 1; \\ 0, & \text{if } j \leq 0. \end{cases} \quad (65)$$

Then, by analogy with (41) and (42), we have ($\mu > 1$)

$$\Delta_2^{(\mu-1)} \binom{1}{j-1} = \gamma_{\mu,j} + \gamma_{\mu,j-1}, \quad (66)$$

$$\gamma_{\mu,j} = \sum_{\substack{1 \leq i_1, \dots, i_{\mu-1} \leq 2 \\ i_1 + \dots + i_{\mu-1} = j-1}} \binom{2}{i_1} \cdots \binom{2}{i_{\mu-1}} = \begin{cases} \binom{\mu-1}{j-\mu} 2^{2\mu-j-1}, & \text{if } \frac{j+1}{2} \leq \mu \leq j; \\ 0 & \text{otherwise.} \end{cases} \quad (67)$$

We introduce the notation

$$X_{\mu,t} = \sum_{i=1}^{s_{0,t}} \binom{u_{i,t}^{(0)}}{\mu}, \quad Y_{\mu,t} = \sum_{i=1}^{s_{1,t}} \binom{u_{i,t}^{(1)}}{\mu}, \quad Z_{\mu,t} = X_{\mu,t} + Y_{\mu,t}.$$

Taking (62)–(64) into account, we can write (61) in the form ($l = 1, \dots, m_\lambda; \lambda = 1, \dots, q$)

$$\begin{aligned} & \sum_{j=1}^n \left(\left(\sum_{t=1}^q k_{\lambda,t} (-1)^{l-1+\sigma} 2^\sigma \gamma_{j,l+\sigma} Z_{j,t} \right) + \left(\sum_{t=1}^q k_{\lambda,t} (-1)^{l-1+\sigma} 2^\sigma \gamma_{j,l-1+\sigma} Y_{j,t} \right) \right. \\ & \quad \left. + (-1)^{m_\lambda} \left(\gamma_{\gamma+1,l+2} \left(\sum_{t=1}^q k_{\lambda,t} \sigma Z_{j,\beta(t)} \right) + \gamma_{j+1,l+1} \left(\sum_{t=1}^q k_{\lambda,t} \sigma Y_{j,\beta(t)} \right) \right) \right) \\ & \quad + \sum_{t=1}^q \sum_{\mu=1}^M \left(\alpha_{\mu,\lambda,l}^{(t)} Z_{\mu,t} + \rho_{\mu,\lambda,l}^{(t)} Y_{\mu,t} \right) = C_{\lambda,l}, \end{aligned} \quad (68)$$

where

$$\begin{aligned} \alpha_{\mu,\lambda,l}^{(t)} &= \sum_{\nu=1}^M 2^{\max(2m_\lambda+1-\nu, \nu)-l} b_{\nu,\lambda,l}^{(t)} \gamma_{\mu+1,\nu+1}, \\ \rho_{\mu,\lambda,l}^{(t)} &= \sum_{\nu=1}^M 2^{\max(2m_\lambda+1-\nu, \nu)-l} b_{\nu,\lambda,l}^{(t)} \gamma_{\mu+1,\nu}. \end{aligned}$$

We set $h(m, \nu) = \max(2m+1-\nu, \nu)$; $\delta(i, j) = 1$, if $i = j$ and $\delta(i, j) = 0$, if $i \neq j$, $U_{\lambda,j} = \sum_{t=1}^q k_{\lambda,t} Z_{j,t}$, $V_{\lambda,j} = \sum_{t=1}^q k_{\lambda,t} Y_{j,t}$ and we introduce the following abbreviation: the symbol $\langle m, \mu \rangle$ will denote a certain 2-adic integral linear combination of $2^m Z_{\mu,t}$ and $2^m Y_{\mu,t}$ ($t = 1, \dots, q$) if $\mu > 0$ and a multiple of 2^m otherwise; here if we insert the index Z , then by $\langle m, \mu \rangle_Z$ we mean that only the $2^m Z_{\mu,t}$ occur in the linear combination (and similarly for the index Y).

We consider the equations (68) for fixed λ . Let $\sigma_{\lambda, m_\lambda}^{(\lambda)} = 0$. Then, by (67),

$$\begin{aligned} & \sum_{(l+1)/2 \leq j \leq n} \gamma_{j,l} U_{\lambda,j} + \sum_{l/2 \leq j \leq n} \gamma_{j,l-1} V_{\lambda,j} \\ &= \langle 0, 0 \rangle + \sum_{\nu=1}^M \left(\sum_{\nu/2 \leq \mu \leq M} \langle h(m_\lambda, \nu) - l + 2\mu - \nu, \mu \rangle_Z \right. \\ & \quad \left. + \sum_{(\nu-1)/2 \leq \mu \leq M} \langle h(m_\lambda, \nu) - l + 2\mu + 1 - \nu, \mu \rangle_Y \right). \end{aligned} \quad (69)$$

Since this system is linear and has a triangular form with respect to the $U_{\lambda,j}$ ($j = 1, \dots, [(m_\lambda + 1)/2]$) and $V_{\lambda,j}$ ($j = 1, \dots, [m_\lambda/2]$), taken alternately in the order of decreasing j , with diagonal entries equal to 1, it follows that the system can be reduced to an equivalent form by expressing these $U_{\lambda,j}$ and $V_{\lambda,j}$ in terms of the others. Using (67)

and induction, starting with $U_{\lambda,[(m_\lambda+1)/2]}$ and $V_{\lambda,[m_\lambda/2]}$, we find that for m_λ odd

$$\begin{aligned}
 U_{\lambda,j} &= \langle 0,0 \rangle + \sum_{(m_\lambda+3)/2 \leq i \leq n} \langle 2i-2j, i \rangle - (m_\lambda-1) \delta\left(j, \frac{m_\lambda+1}{2}\right) V_{\lambda,(m_\lambda+1)/2} \\
 &\quad + \sum_{(m_\lambda+1)/2 \leq i \leq n} \left(1 - \delta\left(j, \frac{m_\lambda+1}{2}\right)\right) \langle 2i+1-2j, i \rangle \\
 &\quad + \delta\left(j, \frac{m_\lambda+1}{2}\right) \left\langle 1, \frac{m_\lambda+1}{2} \right\rangle_Z \\
 &\quad + \sum_{\nu=1}^M \left(\sum_{\nu/2 \leq \mu \leq M} \left(1 - \delta\left(j, \frac{m_\lambda+1}{2}\right) \delta\left(\mu, \frac{m_\lambda+1}{2}\right) \delta(\nu, m_\lambda+1)\right) \right. \\
 &\quad \times \langle h(m_\lambda, \nu) + 1 - 2j + 2\mu - \nu, \mu \rangle \\
 &\quad \left. + \sum_{(\nu-1)/2 \leq \mu \leq M} \langle h(m_\lambda, \nu) + 2 - 2j + 2\mu - \nu, \mu \rangle \right) \\
 &= \langle 0,0 \rangle + \delta\left(j, \frac{m_\lambda-1}{2}\right) \left\langle 3, \frac{m_\lambda-1}{2} \right\rangle \\
 &\quad + \delta\left(j, \frac{m_\lambda+1}{2}\right) \left(\left\langle 1, 0 \right\rangle V_{\lambda,(m_\lambda+1)/2} + \left\langle 1, \frac{m_\lambda+1}{2} \right\rangle_Z \right. \\
 &\quad \left. + \left\langle 1, \frac{m_\lambda-1}{2} \right\rangle + \left\langle 3, \frac{m_\lambda-3}{2} \right\rangle \right) \\
 &\quad + \sum_{\mu=1}^M \left\langle h\left(\frac{m_\lambda+1}{2}, \mu\right) + 1 - j, \mu \right\rangle \quad \left(j = 1, \dots, \frac{m_\lambda+1}{2} \right), \tag{70}
 \end{aligned}$$

$$\begin{aligned}
 V_{\lambda,j} &= \langle 0,0 \rangle + \sum_{(m_\lambda+3)/2 \leq i \leq n} \langle 2i-2j-1, i \rangle + \sum_{(m_\lambda+1)/2 \leq i \leq n} \langle 2i-2j, i \rangle \\
 &\quad + \sum_{\nu=1}^M \left(\sum_{\nu/2 \leq \mu \leq M} \langle h(m_\lambda, \nu) - 2j + 2\mu - \nu, \mu \rangle \right. \\
 &\quad \left. + \sum_{(\nu-1)/2 \leq \mu \leq M} \langle h(m_\lambda, \nu) + 1 - 2j + 2\mu - \nu, \mu \rangle \right) \\
 &= \langle 0,0 \rangle + \delta\left(j, \frac{m_\lambda-3}{2}\right) \left\langle 4, \frac{m_\lambda-1}{2} \right\rangle \\
 &\quad + \delta\left(j, \frac{m_\lambda-1}{2}\right) \left(\left\langle 2, \frac{m_\lambda+1}{2} \right\rangle + \left\langle 2, \frac{m_\lambda-1}{2} \right\rangle + \left\langle 4, \frac{m_\lambda-3}{2} \right\rangle \right) \\
 &\quad + \sum_{\mu=1}^M \left\langle h\left(\frac{m_\lambda+1}{2}, \mu\right) + 1 - j, \mu \right\rangle \quad \left(j = 1, \dots, \frac{m_\lambda-1}{2} \right), \tag{71}
 \end{aligned}$$

and for m_λ even

$$\begin{aligned}
 U_{\lambda,j} &= \langle 0,0 \rangle + \sum_{m_\lambda/2+1 \leq i \leq n} \langle 2i-2j, i \rangle \\
 &\quad + \sum_{\nu=1}^M \left(\sum_{\nu/2 \leq \mu \leq M} \langle h(m_\lambda, \nu) + 1 - 2j + 2\mu - \nu, \mu \rangle \right. \\
 &\quad \left. + \sum_{(\nu-1)/2 \leq \mu \leq M} \langle h(m_\lambda, \nu) + 2 - 2j + 2\mu - \nu, \mu \rangle \right) \\
 &= \langle 0,0 \rangle + \delta\left(j, \frac{m_\lambda}{2} - 1\right) \left(\left\langle 4, \frac{m_\lambda}{2} \right\rangle + \left\langle 5, \frac{m_\lambda - 1}{2} \right\rangle \right) \\
 &\quad + \delta\left(j, \frac{m_\lambda}{2}\right) \left(\left\langle 2, \frac{m_\lambda}{2} + 1 \right\rangle + \left\langle 2, \frac{m_\lambda}{2} \right\rangle + \left\langle 3, \frac{m_\lambda}{2} - 1 \right\rangle + \left\langle 5, \frac{m_\lambda}{2} - 2 \right\rangle \right) \\
 &\quad + \sum_{\mu=1}^M \left\langle h\left(\frac{m_\lambda}{2} + 1, \mu\right) + 1 - j, \mu \right\rangle \quad \left(j = 1, \dots, \frac{m_\lambda}{2} \right), \tag{72}
 \end{aligned}$$

$$\begin{aligned}
 V_{\lambda,j} &= \langle 0,0 \rangle + \delta\left(j, \frac{m_\lambda}{2}\right) \left(-m_\lambda U_{\lambda, m_\lambda/2+1} + \left\langle 1, \frac{m_\lambda}{2} \right\rangle_Z \right) \\
 &\quad + \sum_{m_\lambda/2+1 \leq i \leq n} \langle \max(2, 2i-2j-1), i \rangle \\
 &\quad + \sum_{\nu=1}^M \left(\sum_{\nu/2 \leq \mu \leq M} \left(1 - \delta\left(j, \frac{m_\lambda}{2}\right) \delta\left(j, \frac{m_\lambda}{2}\right) \delta(\nu, m_\lambda) \right) \right. \\
 &\quad \times \langle h(m_\lambda, \nu) - 2j + 2\mu - \nu, \mu \rangle \\
 &\quad \left. + \sum_{(\nu-1)/2 \leq \mu \leq M} \langle h(m_\lambda, \nu) + 1 - 2j + 2\mu - \nu, \mu \rangle \right) \\
 &= \langle 0,0 \rangle + \delta\left(j, \frac{m_\lambda}{2} - 2\right) \left(\left\langle 5, \frac{m_\lambda}{2} \right\rangle + \left\langle 6, \frac{m_\lambda}{2} - 1 \right\rangle \right) \\
 &\quad + \delta\left(j, \frac{m_\lambda}{2} - 1\right) \left(\left\langle 3, \frac{m_\lambda}{2} \right\rangle + \left\langle 4, \frac{m_\lambda}{2} - 1 \right\rangle + \left\langle 6, \frac{m_\lambda}{2} - 2 \right\rangle \right) \\
 &\quad + \delta\left(j, \frac{m_\lambda}{2}\right) \left(-m_\lambda U_{\lambda, m_\lambda/2+1} + \left\langle 2, \frac{m_\lambda}{2} + 1 \right\rangle + \left\langle 1, \frac{m_\lambda}{2} \right\rangle_Z + \left\langle 2, \frac{m_\lambda}{2} - 1 \right\rangle \right. \\
 &\quad \left. + \left\langle 4, \frac{m_\lambda}{2} - 2 \right\rangle + \left\langle 6, \frac{m_\lambda}{2} - 3 \right\rangle \right) \\
 &\quad + \sum_{\mu=1}^M \left\langle h\left(\frac{m_\lambda}{2} + 1, \mu\right) + 1 - j, \mu \right\rangle \quad \left(j = 1, \dots, \frac{m_\lambda}{2} \right). \tag{73}
 \end{aligned}$$

Let $\sigma_{\lambda, m_\lambda}^{(\lambda)} = 1$. Then m_λ is odd, and, since one of the numbers t_r or $\beta(t_r)$ is $2r - 1$ and the other is $2r$ when $m_0 > 1$, it follows from (68) that for $l = m_\lambda$

$$\begin{aligned} & \sum_{j=1}^n \sum_{1 \leq r \leq q/2} \left(2k_{\lambda, t_r} (\gamma_{j, m_\lambda+1} Z_{j, t_r} + \gamma_{j, m_\lambda} Y_{j, t_r}) + k_{\lambda, \beta(t_r)} (\gamma_{j, m_\lambda} Z_{j, \beta(t_r)} + \gamma_{j, m_\lambda-1} Y_{j, \beta(t_r)}) \right. \\ & \quad \left. + k_{\lambda, t_r} (\gamma_{j+1, m_\lambda+2} Z_{j, \beta(t_r)} + \gamma_{j+1, m_\lambda+1} Y_{j, \beta(t_r)}) \right) \\ & = \langle 0, 0 \rangle + \left\langle 1, \frac{m_\lambda + 1}{2} \right\rangle_Z + \left\langle 1, \frac{m_\lambda - 1}{2} \right\rangle + \left\langle 3, \frac{m_\lambda - 3}{2} \right\rangle \\ & \quad + \sum_{\mu=1}^M \left\langle h \left(\frac{m_\lambda + 1}{2}, \mu \right) + 1 - \frac{m_\lambda + 1}{2}, \mu \right\rangle, \end{aligned}$$

which, using (67) and the congruence $k_{\lambda, \beta(t_r)} \equiv 0 \pmod{2}$ (see the conditions on $\sigma_{\lambda, j}^{(l)}$), we rewrite in the form

$$\begin{aligned} & \sum_{1 \leq r \leq q/2} k_{\lambda, t_r} (2Y_{(m_\lambda+1)/2, t_r} + Z_{(m_\lambda+1)/2, \beta(t_r)} + (m_\lambda + 1)Y_{(m_\lambda+1)/2, \beta(t_r)}) \\ & = \langle 0, 0 \rangle + \left\langle 1, \frac{m_\lambda + 1}{2} \right\rangle_Z + \left\langle 1, \frac{m_\lambda - 1}{2} \right\rangle \\ & \quad + \left\langle 3, \frac{m_\lambda - 3}{2} \right\rangle + \sum_{\mu=1}^M \left\langle h \left(\frac{m_\lambda + 1}{2}, \mu \right) + 1 - \frac{m_\lambda + 1}{2}, \mu \right\rangle. \end{aligned} \quad (74)$$

Since $\sigma_{\lambda, l}^{(l)} = 0$ for $l < m_\lambda$, it follows that the remaining equations in (68) can be written in the form (69). From the conditions $\sigma_{\lambda, m_\lambda}^{(\lambda)} = 1$ we have $m_\lambda = m_{\lambda+1}$ and $\sigma_{\lambda+1, m_{\lambda+1}}^{(\lambda+1)} = 0$. Consequently, we have (70) and (71) for $\lambda + 1$. In particular, (70) with $j = (m_\lambda + 1)/2$ and λ replaced by $\lambda + 1$ gives

$$\begin{aligned} U_{\lambda+1, (m_\lambda+1)/2} & = \sum_{t=1}^q k_{\lambda+1, t} Z_{(m_\lambda+1)/2, t} \\ & = \langle 0, 0 \rangle + \langle 1, 0 \rangle V_{\lambda+1, (m_\lambda+1)/2} + \left\langle 1, \frac{m_\lambda + 1}{2} \right\rangle_Z + \left\langle 1, \frac{m_\lambda - 1}{2} \right\rangle \\ & \quad + \left\langle 3, \frac{m_\lambda - 3}{2} \right\rangle + \sum_{\mu=1}^M \left\langle h \left(\frac{m_\lambda + 1}{2}, \mu \right) + 1 - \frac{m_\lambda + 1}{2}, \mu \right\rangle. \end{aligned} \quad (75)$$

Using the congruences (see the conditions on $\sigma_{\lambda, j}^{(l)}$)

$$k_{\lambda, \beta(t_r)} \equiv 0 \pmod{2}, \quad k_{\lambda, t_r} \equiv k_{\lambda+1, 2r-1} \equiv k_{\lambda, 2r} \pmod{2}$$

and (75), we replace (74) by the equation

$$\begin{aligned} U_{\lambda, (m_\lambda+1)/2} & = \langle 0, 0 \rangle + \langle 1, 0 \rangle V_{\lambda, (m_\lambda+1)/2} + \langle 1, 0 \rangle V_{\lambda+1, (m_\lambda+1)/2} \\ & \quad + \left\langle 1, \frac{m_\lambda + 1}{2} \right\rangle_Z + \left\langle 1, \frac{m_\lambda - 1}{2} \right\rangle + \left\langle 3, \frac{m_\lambda - 3}{2} \right\rangle \\ & \quad + \sum_{\mu=1}^M \left\langle h \left(\frac{m_\lambda + 1}{2}, \mu \right) + 1 - \frac{m_\lambda + 1}{2}, \mu \right\rangle. \end{aligned} \quad (76)$$

We now transform (70)–(73) and (76). We set

$$m' = \min\left(\left[\frac{m_1}{2}\right], \dots, \left[\frac{m_q}{2}\right]\right), \quad m'' = \max\left(\left[\frac{m_1+1}{2}\right], \dots, \left[\frac{m_q+1}{2}\right]\right).$$

From the inequalities $|m_i - m_j| \leq 1$ ($i, j = 1, \dots, q$) it follows that $0 \leq m'' - m' \leq 1$ and, for m_λ odd, $m'' = (m_\lambda + 1)/2 \neq m'$. Since $\det((k_{\lambda,t})_{t=1,\dots,q}^{\lambda=1,\dots,q}) \not\equiv 0 \pmod{2}$, it follows from (70)–(73) that for each j , $1 \leq j \leq m'$, we can express $Z_{j,t}$ and $Y_{j,t}$ ($t = 1, \dots, q$) in the form

$$\begin{aligned} Z_{j,t} = & \langle 0, 0 \rangle + \delta(j, m')(\langle 2, m' + 1 \rangle + \langle 2, m' \rangle + \langle 3, m' - 1 \rangle + \langle 5, m' - 2 \rangle) \\ & + \delta(j, m' - 1)(\langle 4, m' \rangle + \langle 5, m' \rangle + 1) + \sum_{\mu=1}^M \langle h(m'', \mu) + 1 - j, \mu \rangle, \end{aligned} \quad (77)$$

$$\begin{aligned} Y_{j,t} = & \langle 0, 0 \rangle + \delta(j, m')(\langle 1, m' + 1 \rangle + \langle 1, m' \rangle + \langle 2, m' - 1 \rangle \\ & + \langle 4, m' - 2 \rangle + \langle 6, m' - 3 \rangle) \\ & + \delta(j, m' - 1)(\langle 3, m' \rangle + \langle 4, m' - 1 \rangle + \langle 6, m' - 2 \rangle) \\ & + \delta(j, m' - 2)(\langle 5, m' \rangle + \langle 6, m' - 1 \rangle) + \sum_{\mu=1}^M \langle h(m'', \mu) + 1 - j, \mu \rangle. \end{aligned} \quad (78)$$

In (77) and (78) we now get rid of the terms of the form $\langle k, i \rangle$ with $m' - 3 \leq i \leq m'$ by putting them into $\langle 0, 0 \rangle$ and into the sum Σ by means of several substitutions of the expressions $Z_{i,t}$ and $Y_{i,t}$ in $\langle k, i \rangle$. We then obtain the equations ($j = 1, \dots, m'$; $t = 1, \dots, q$)

$$\begin{aligned} Z_{j,t} = & \langle 0, 0 \rangle + \delta(j, m')\langle 2, m' + 1 \rangle + \sum_{\mu=1}^M \langle h(m'', \mu) + 1 - j, \mu \rangle, \\ Y_{j,t} = & \langle 0, 0 \rangle + \delta(j, m')\langle 1, m' + 1 \rangle + \sum_{\mu=1}^M \langle h(m'', \mu) + 1 - j, \mu \rangle. \end{aligned} \quad (79)$$

We substitute these expressions into (70)–(73) and (76). Then (70) and (71) become the equations

$$\begin{aligned} U_{\lambda,j} = & \langle 0, 0 \rangle + \delta\left(j, \frac{m_\lambda + 1}{2}\right) \left(\langle 1, 0 \rangle V_{\lambda, (m_\lambda + 1)/2} + \left\langle 1, \frac{m_\lambda + 1}{2} \right\rangle_Z \right) \\ & + \sum_{\mu=1}^M \left\langle h\left(\frac{m_\lambda + 1}{2}, \mu\right) + 1 - j, \mu \right\rangle \quad \left(j = 1, \dots, \frac{m_\lambda + 1}{2} \right), \end{aligned} \quad (80)$$

$$\begin{aligned} V_{\lambda,j} = & \langle 0, 0 \rangle + \delta\left(j, \frac{m_\lambda - 1}{2}\right) \left\langle 2, \frac{m_\lambda + 1}{2} \right\rangle \\ & + \sum_{\mu=1}^M \left\langle h\left(\frac{m_\lambda + 1}{2}, \mu\right) + 1 - j, \mu \right\rangle \quad \left(j = 1, \dots, \frac{m_\lambda - 1}{2} \right), \end{aligned} \quad (81)$$

equation (76) becomes

$$\begin{aligned} U_{\lambda, (m_\lambda + 1)/2} = & \langle 0, 0 \rangle + \langle 1, 0 \rangle V_{\lambda, (m_\lambda + 1)/2} + \langle 1, 0 \rangle V_{\lambda + 1, (m_\lambda + 1)/2} \\ & + \left\langle 1, \frac{m_\lambda + 1}{2} \right\rangle_Z + \sum_{\mu=1}^M \left\langle h\left(\frac{m_\lambda + 1}{2}, \mu\right) + 1 - \frac{m_\lambda + 1}{2}, \mu \right\rangle \end{aligned} \quad (82)$$

and (72) and (73) become ($j = 1, \dots, m_\lambda/2$)

$$U_{\lambda,j} = \langle 0, 0 \rangle + \sum_{\mu=1}^M \left\langle h\left(\frac{m_\lambda}{2}, \mu\right) + 1 - j, \mu \right\rangle, \quad (83)$$

$$\begin{aligned} V_{\lambda,j} = \langle 0, 0 \rangle + \delta\left(j, \frac{m_\lambda}{2}\right) & \left(-m_\lambda U_{\lambda, m_\lambda/2+1} + \left\langle 1, \frac{m_\lambda}{2} \right\rangle_Z \right) \\ & + \sum_{\mu=1}^M \left\langle h\left(\frac{m_\lambda}{2}, \mu\right) + 1 - j, \mu \right\rangle. \end{aligned} \quad (84)$$

The next transformations consist in reducing (80)–(84) to the form (52). We first consider (80) and (81). We set $V'_{\lambda,j} = \sum_{t=1}^q k_{\lambda,t} X_{j,t}$. Since $U_{\lambda,j} = V_{\lambda,j} + V'_{\lambda,j}$, it follows from (80) and (81) that for $j = 1, \dots, (m_\lambda - 1)/2$ there are equations for $V'_{\lambda,j}$ which are analogous to (81):

$$\begin{aligned} V'_{\lambda,j} = \langle 0, 0 \rangle + \delta\left(j, \frac{m_\lambda - 1}{2}\right) & \left\langle 2, \frac{m_\lambda + 1}{2} \right\rangle \\ & + \sum_{\mu=1}^M \left\langle h\left(\frac{m_\lambda + 1}{2}, \mu\right) + 1 - j, \mu \right\rangle. \end{aligned} \quad (85)$$

We write (80) with $j = (m_\lambda + 1)/2$ in the form

$$\begin{aligned} U_{\lambda, (m_\lambda + 1)/2} + 2aV_{\lambda, (m_\lambda + 1)/2} + \sum_{t=1}^q 2b_t Z_{(m_\lambda + 1)/2, t} \\ = \langle 0, 0 \rangle + \sum_{\mu=1}^M \left\langle h\left(\frac{m_\lambda + 1}{2}, \mu\right) + 1 - \frac{m_\lambda + 1}{2}, \mu \right\rangle, \end{aligned} \quad (86)$$

where a and b_t are 2-adic integers. We set

$$U_{\lambda,j}^* = U_{\lambda,j} + 2aV_{\lambda,j} + \sum_{t=1}^q 2b_t Z_{j,t}. \quad (87)$$

From (80) and (81), if we use (86), (87), and (79), we obtain ($j = 1, \dots, (m_\lambda + 1)/2$)

$$U_{\lambda,j}^* = \langle 0, 0 \rangle + \sum_{\mu=1}^M \left\langle h\left(\frac{m_\lambda + 1}{2}, \mu\right) + 1 - j, \mu \right\rangle. \quad (88)$$

We bring the equations in (88) together with those in (81) if $s_{1,\lambda} = H((m_\lambda - 1)/2)$ or in (85) if $s_{0,\lambda} = H((m_\lambda - 1)/2)$, and we write them in the form ($j = 1, \dots, (m_\lambda - 1)/2$):

$$V_{\lambda,j} \text{ or } V'_{\lambda,j} = \langle 0, 0 \rangle + \sum_{\mu=1}^M \left\langle h\left(\frac{m_\lambda - 1}{2}, \mu\right) + 1 - j, \mu \right\rangle. \quad (89)$$

The equations in (82) and (80), (81) with $j = 1, \dots, (m_\lambda - 1)/2$, which correspond to the case $\sigma_{\lambda, m_\lambda}^{(\lambda)} = 1$, can also be reduced to the form (88), (89) with the same conditions on $s_{1,\lambda}$ and $s_{0,\lambda}$. To do this we rewrite (82) in the form

$$\begin{aligned} U_{\lambda, (m_\lambda + 1)/2} + 2a'V_{\lambda, (m_\lambda + 1)/2} + 2a''V_{\lambda+1, (m_\lambda + 1)/2} + \sum_{t=1}^q 2b'_t Z_{(m_\lambda + 1)/2, t} \\ = \langle 0, 0 \rangle + \sum_{\mu=1}^M \left\langle h\left(\frac{m_\lambda + 1}{2}, \mu\right) + 1 - \frac{m_\lambda + 1}{2}, \mu \right\rangle, \end{aligned}$$

where a' , a'' , and b'_t are 2-adic integers, and we set

$$U_{\lambda,j}^* = U_{\lambda,j} + 2a'V_{\lambda,j} + 2a''V_{\lambda+1,j} + \sum_{t=1}^q 2b'_t Z_{j,t}. \quad (90)$$

Then, just as in the previous case, we obtain equations of the type (88), (89).

We now consider (83) and (84). We write (84) with $j = m_\lambda/2$ in the form

$$\begin{aligned} V_{\lambda, m_\lambda/2} + \sum_{t=1}^q 2d_t Z_{m_\lambda/2, t} &= 4\left(\frac{m_\lambda}{4}\right) \left(U_{\lambda, m_\lambda/2+1} + \sum_{t=1}^q 2d_t Z_{m_\lambda/2+1, t} \right) \\ &+ \langle 0, 0 \rangle + \sum_{\mu=1}^M \left\langle h\left(\frac{m_\lambda}{2}, \mu\right) + 1 - \frac{m_\lambda}{2}, \mu \right\rangle, \end{aligned}$$

where the d_t are 2-adic integers, and we set

$$V_{\lambda,j}^* = V_{\lambda,j} + \sum_{t=1}^q 2d_t Z_{j,t}. \quad (91)$$

Using (79), from (84) we obtain the equations ($j = 1, \dots, m_\lambda/2$)

$$\begin{aligned} V_{\lambda,j}^* - \delta\left(j, \frac{m_\lambda}{2}\right) 4\left(\frac{m_\lambda}{4}\right) \left(U_{\lambda, j+1} + \sum_{t=1}^q 2d_t Z_{j+1, t} \right) \\ = \langle 0, 0 \rangle + \sum_{\mu=1}^M \left\langle h\left(\frac{m_\lambda}{2}, \mu\right) + 1 - j, \mu \right\rangle. \end{aligned} \quad (92)$$

We shall not change the equations in (83).

As a result of these linear transformations of the system (68) we have obtained a system of equations consisting of the equations in (88) and (89) for each λ with m_λ odd and the equations in (92) and (93) for each λ with m_λ even. From (87), (90), (91), and the condition $\det((k_{\lambda,t})_{t=1,\dots,q}^{\lambda=1,\dots,q}) \not\equiv 0 \pmod{2}$ it follows that this system has the same number $m_1 + \dots + m_q$ of linearly independent (over the field of 2-adic numbers) equations as system (68); thus, it is equivalent to (68). We introduce new notation for the variables $u_{i,t}^{(e)}$. If m_t is even, then we denote $u_{i,t}^{(e)}$ by $y_{i,2t-\varepsilon}$, and if m_t is odd, then we denote whichever of the variables $u_{i,t}^{(e)}$ ($\varepsilon = 0, 1$) has $s_{\varepsilon,t} = H((m_t - 1)/2)$ by the symbol $y_{i,2t-1}$, and we denote the other variable by $y_{i,2t}$. We set $m_{2t-1}^* = [m_t/2]$ and $m_{2t}^* = [(m_t + 1)/2]$; for odd m_t we set $\kappa_t^* = 2t - 1$, if $s_{1,t} = H((m_t - 1)/2)$ and $\kappa_t^* = 2t$ if $s_{1,t} = H((m_t + 1)/2)$, and for even m_t we set $\kappa_t^* = 2t - 1$. In this new notation our system takes the form

$$\begin{aligned} \sum_{r=1}^{2q} \sum_{i=1}^{H(m_r^*)} k_{\nu,r}^* \left(\binom{y_{i,r}}{j} - 2\sigma_{\nu,j}^{(r)} \left(\binom{y_{i,r}}{j+1} + \binom{y_{i,\beta(r)}}{j+1} \right) \right) \\ = \langle 0, 0 \rangle + \sum_{\mu=1}^M \left\langle h(m_\nu^*, \mu) 1 - j, \mu \right\rangle \quad (j = 1, \dots, m_\nu^*; \nu = 1, \dots, 2q), \end{aligned} \quad (93)$$

where $\langle l, \mu \rangle$ is a 2-adic integral linear combination of

$$2^l \sum_{1 \leq i \leq H(m_r^*)} \binom{y_{i,r}}{\mu} \quad (r = 1, \dots, 2q); \quad \beta(r) = 4 \left\lfloor \frac{r+1}{2} \right\rfloor - r - 1;$$

the numbers $k_{\nu,r}^*$ satisfy the following conditions: $k_{\nu,r}^* \equiv 0 \pmod{2}$, if $r > \nu$; $k_{\nu,r}^* \equiv k_{\lambda,\lambda} \equiv 1 \pmod{2}$, if $\nu = r = 2\lambda - 1$ or 2λ and m_λ is odd; for m_λ even $k_{\nu,r}^* = k_{\lambda, [(r+1)/2]}$, if $\nu = 2\lambda$, $k_{\nu,r}^* = k_{\lambda,t} + 2d_t$, if $\nu = 2\lambda - 1$, $r = \kappa_t^*$, and $k_{\nu,r}^* = 2d_t$ if $\nu = 2\lambda - 1$,

$r = \beta(\kappa_t^*)$, and

$$\sigma_{\nu,j}^{(r)} = \begin{cases} 1, & \text{if } \nu = 2\lambda - 1, m_\lambda \text{ is even, } r = \kappa_t^* \text{ for some } t \in [1, q], \\ m_\lambda/2 \text{ is odd, } j = m_\lambda/2 = m_{2\lambda-1}^* = m_{2\lambda}^*; \\ 0, & \text{otherwise.} \end{cases}$$

Let $m' = \min(m_1^*, \dots, m_{2q}^*) > 0$. Then system (93) has the form (52) with $2q$ in place of q . If $m' = 0$, then it is necessary to discard the empty equations and the variables which occur formally in (93) with $m_\nu^* = 0$ and $m_r^* = 0$. After this, if we make a suitable reindexing, system (93) takes the form (52) with n in place of q .

Since $m'' = \max(m_1^*, \dots, m_{2q}^*) = [(m_0 + 1)/2] < m_0$, it follows by the induction assumption that $H(m_r^*)$ variables $y_{i,r}$ for each r are sufficient for solvability of system (93) in 2-adic integers $y_{i,r}$ which for each $r = 1, \dots, 2q$ contain a rank- m_r^* lifting tuple (mod 2); this implies that (61) is solvable in 2-adic integers $u_{i,t}^{(\epsilon)}$ which for each t and ϵ contain a lifting tuple (mod 2) of rank $[m_t/2]$ if $s_{\epsilon,t} = H([m_t/2])$, or of rank $[(m_t + 1)/2]$ if $s_{\epsilon,t} = H([(m_t + 1)/2])$. The proposition is proved.

For $q = 1$ we see that Proposition 2 implies

COROLLARY.

$$\bar{r}(n, 2) \leq \sum_{0 \leq k \leq [\ln n / \ln 2]} 2^k (2^{\lfloor n/2^k \rfloor} - 1).$$

THEOREM. The following bound holds for $n \geq 12$:

$$r(n) \leq \sum_{0 \leq k \leq [\ln n / \ln 2]} 2^k (2^{\lfloor n/2^k \rfloor} - 1).$$

PROOF. From Lemma 11, Proposition 1, the corollary to Proposition 2, the corollary to Lemma 12, and Lemma 15 we find that for $n \geq 12$

$$\begin{aligned} r(n) &\leq \max \left(n^2 (4 \ln n + 2 \ln \ln n + 9), 2en^2 \ln n + n, H(n), \max_{3 \leq p \leq n} p^{(n-1)/(p-1)+2} \right) \\ &\leq \max \left(H(n), \max_{3 \leq p \leq n} p^{(n-1)/(p-1)+2} \right). \end{aligned} \quad (94)$$

Since the function $F(x) = ((n-1)/(x-1) + 2) \ln x$ has a negative derivative for $5 \leq x \leq (n+1)/2$:

$$F'(x) = -\frac{n-1}{(x-1)^2} \ln x + \frac{1}{x} \left(\frac{n-1}{x-1} + 2 \right) \leq \frac{n-1}{(x-1)^2} \left(\frac{2(x-1)}{x} - \ln x \right) < 0,$$

it follows that

$$\max_{3 \leq p \leq n} p^{(n-1)/(p-1)+2} \leq \max \left(3^{(n+3)/2}, 5^{(n+7)/4}, \max_{(n+1)/2 < p \leq n} p^{(n-1)/(p-1)+2} \right).$$

Furthermore, for $n \geq 12$ we have

$$3^{(n+3)/2} < 2^{(8/5)(n+3)/2} \leq 2^n, \quad 5^{(n+7)/4} < 2^{(7/3)(n+7)/4} < 2^n,$$

and for $n \geq 16$ we have

$$\max_{(n+1)/2 < p \leq n} p^{(n-1)/(p-1)+2} \leq n^{4-2/n} < 2^n.$$

The inequality

$$\max_{(n+1)/2 < p \leq n} p^{(n-1)/(p-1)+2} \leq 2^n$$

for $12 \leq n \leq 15$ follows from the inequalities

$$7^{(n-1)/6+2} < 2^{3((n-1)/6+2)} < 2^n, \quad 11^{(n-1)/10+2} < 2^{(7/2)((n-1)/10+2)} < 2^n, \\ 13^{(n-1)/12+2} < 2^{4((n-1)/12+2)} < 2^n.$$

From (94) we now obtain $r(n) \leq H(n)$. The theorem is proved.

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Received 20/APR/85

BIBLIOGRAPHY

1. G. I. Arkhipov, *On the value of the singular series in the Hilbert-Kamke problem*, Dokl. Akad. Nauk SSSR **259** (1981), 265–267; English transl. in Soviet Math. Dokl. **24** (1981).
2. ———, *On the Hilbert-Kamke problem*, Izv. Akad. Nauk SSSR Ser. Mat. **48** (1984), 3–52; English transl. in Math. USSR Izv. **24** (1985).
3. E. Kamke, *Verallgemeinerung des Waring-Hilbertschen Satzes*, Math. Ann. **83** (1921), 85–112.
4. David Hilbert, *Beweis für den Darstellbarkeit der ganzen Zahlen durch eine feste Anzahlter Potenzen*, Math. Ann. **67** (1909), 281–300.
5. I. M. Vinogradov, *A new method in analytic number theory*, Trudy Mat. Inst. Steklov. **10** (1937). (Russian) [Earlier version of [6] and [7].]
6. ———, *The method of trigonometric sums in the theory of numbers*, Trudy Mat. Inst. Steklov. **23** (1947); English transl., Interscience, 1954.
7. ———, *The method of trigonometric sums in number theory*, latest ed., “Nauka”, Moscow, 1980; English transl. in his *Selected works*, Springer-Verlag, 1985.
8. K. K. Mardzhanishvili, *Sur la représentation simultanée de n nombres par des sommes des puissances complètes*, Izv. Akad. Nauk SSSR Otd. Mat. Estestv. Nauk **1937** [= Ser. Mat. **1** (1937)], 609–631. (Russian; French summary)
9. ———, *On some nonlinear systems of equations in integers*, Mat. Sb. **33** (75) (1953), 639–675. (Russian)
10. G. V. Emel'yanov, *On a system of Diophantine equations*, Leningrad. Gos. Univ. Uchen. Zap. **137** (Ser. Mat. Nauk **19**) (1950), 3–39. (Russian)
11. Hua Loo-Keng, *Additive theory of prime numbers*, Trudy Mat. Inst. Steklov. **22** (1947); English transl. of rev. Chinese ed., Amer. Math. Soc., Providence, R. I., 1965.
12. A. O. Gel'fond, *The calculus of finite differences*, 3rd ed., “Nauka”, Moscow, 1967; English transl., Hindustan, Delhi, 1971.
13. K. K. Mardzhanishvili, *On a singular series*, Trudy Mat. Inst. Steklov. **142** (1976), 174–181; English transl. in Proc. Steklov Inst. Math. **1979**, no. 3 (142).
14. L. J. Mordell, *On a sum analogous to a Gauss's sum*, Quart. J. Math. Oxford Ser. **3** (1932), 161–167.
15. A. A. Karatsuba, *On a certain system of congruences*, Mat. Zametki **19** (1976), 389–392; English transl. in Math. Notes **19** (1976).
16. Yu. V. Linnik, *Some remarks on estimates of trigonometric sums*, Uspekhi Mat. Nauk **14** (1959), no. 3 (87), 153–160; English transl. in Amer. Math. Soc. Transl. (2) **43** (1964).
17. André Weil, *On some exponential sums*, Proc. Nat. Acad. Sci. U.S.A. **34** (1948), 204–207.
18. Z. I. Borevich and I. R. Shafarevich, *Number theory*, 2nd rev. ed., “Nauka”, Moscow, 1972; English transl. of 1st ed., Academic Press, 1966.
19. Hua Loo-Keng, *On a generalized Waring problem. II*, J. Chinese Math. Soc. **2** (1940), 175–191.
20. V. I. Nechaev, *An estimate of the complete rational trigonometric sum*, Mat. Zametki **17** (1975), 839–849; English transl. in Math. Notes **17** (1975).

Translated by N. KOBLITZ